DANMARKS OG GRØNLANDS GEOLOGISKE UNDERSØGELSE **RAPPORT 2022/12**

Geochemical Data Collection in LOOP6

MapField, 2019-2020

Hyojin Kim, Rasmus Jakobsen, Jens Aamand, Ingelise Møller & Birgitte Hansen

GEOLOGICAL SURVEY OF DENMARK AND GREENLAND
DANISH MINISTRY OF CLIMATE, ENERGY AND UTILITIES

G E U S

DANMARKS OG GRØNLANDS GEOLOGISKE UNDERSØGELSE **RAPPORT 2022/12**

Geochemical Data Collection in LOOP6

MapField, 2019-2020

Results from The Innovation Fund Denmark project:
MapField – Field-scale mapping for targeted Nregulation
and management (8855-00025B)

Hyojin Kim, Rasmus Jakobsen, Jens Aamand, Ingelise Møller & Birgitte Hansen

Confidential report

Copy No.

Released 30.09.2024

GEOLOGICAL SURVEY OF DENMARK AND GREENLAND
DANISH MINISTRY OF CLIMATE, ENERGY AND UTILITIES

G E U S

Geochemical data collection in LOOP6 MapField, 2019-2020

Hyojin Kim, Rasmus Jakobsen, Jens Aamand, Ingelise Møller & Birgitte Hansen

G E U S

DE NATIONALE GEOLOGISKE UNDERSØGELSER FOR DANMARK OG GRØNLAND

KLIMA- OG ENERGIMINISTERIET

Data report 2019-2020 MapField

Authors: Hyojin Kim, Rasmus Jakobsen, Jens Aamand, Ingelise Møller & Birgitte Hansen

Table of content

1. Introduction **4**

2. Methods **5**

2.1. Field campaign planning 5
2.2. Identified hypotheses 5
2.3. Borehole drilling and sample collection 6
2.4. Gas sampling well installation 6
2.5. Water chemistry analysis 6
2.6. Sediment chemistry analysis 6
2.7. Nitrate reduction rate measurements and experiments 7
2.8. Gas chemistry analysis 9
2.9. Lithology description 9
2.10. Redox probe and IP measurements 9

3. Results **11**

3.1. Existing data 11
3.2. Collected samples 15
3.3. Overview of gas sample collection 16
3.4. Interpretation of redox zones 16
3.5. Overview of the MapField chemical analyses 17
3.6. Results from the rate experiments 17

Appendices **17**

Appendix 1: Lithological descriptions of the MapField samples 18
Appendix 2: Water chemistry of existing data (LOOP, GRUMO and others) 18
Appendix 3: Water and sediment chemistry collected in the MapField project 18
Appendix 4: Well panels illustrating all the collected parameters 18

1. Introduction

This report summarizes existing data and results of the field campaigns carried out in 2019 to collect geochemistry data for the MapField project. The geochemistry data includes groundwater chemistry, sediment chemistry, and nitrate reduction rates. We drilled boreholes to collect groundwater and sediment samples. The primary objectives of the geochemistry data collection are to capture transport and evolution of nitrate in the subsurface at the field scale and to quantify the rates of nitrate reduction. This geochemistry data will be input data for an integrated hydro-geochemistry modelling. Here, we describe the methods and results.

2. Methods

2.1. Field campaign planning

To select the most important drilling sites of the study catchment, preliminary characterization of the subsurface structure and nitrate transport and fate was done using existing data. The existing data encompass 1) tTEM and geological interpretation of the tTEM model; 2) digital terrain model; 3) water chemistry (pore water, groundwater, and stream water chemistry) focusing on nitrate; 4) groundwater table; and 5) redox zones interpreted from sediment colors.

2.2. Identified hypotheses

Based on this preliminary analysis, hypotheses for the evolution of nitrate in the subsurface were developed and sampling points for N reduction rates, and detailed water chemistry profiling were selected. Both hypotheses related to optimization and iterative creative processes were identified:

Optimization process:

1. Using 2.5 days for screening the redox conditions and IP (Induced polarization) with the redox-IP-geoprobe is optimal for planning the 3-days drilling campaign for sediment, water and gas sampling

Iterative creative process:

1. There is a significant correlation between redox state and IP at the redox interfaces due to facies change from Fe^{3+} to Fe^{2+}.
2. N reduction in the outwash plain with peat soil is controlled by oxidation of organic matter.
3. N reduction in the outwash plain is also controlled by fast oxidation of pyrite in other areas.
4. Fast transport of N through drains controls the stream transport of N at the boundaries of the Saale hilly landscape and the outwash plain.
5. N reduction in the unsaturated zone plays a measurable role in the total N reduction.
6. Exploring different laboratory methods for N reduction rate determination e.g. with and without the use of acetylene.

2.3. Borehole drilling and sample collection

In September 2019, to screen the redox interface of the selected sampling points, a redox probe survey was carried out. The groundwater table was measured as well. Two weeks after the redox probe survey, based on the screeing results, a Geoprobe direct push method was used to collect core samples less than 1 meter from the redox probe points. The drilling was performed by Ejlskov. The core samples were cut and wrapped with aluminum tape in the field for pore-water chemistry (9cm-long), nitrate reduction rates (25cm-long), geochemistry analysis (varying length), and lithological description (varying length). Except the lithology samples, all the core samples were stored in a cooling box in the field.

2.4. Gas sampling well installation

During the drilling campaign, gas sampling wells were installed in selected locations among the geochemistry sampling points. Plastic tubing was cut for a desired length and about 20-30 cm of the bottom end was slotted to allow gas influx. The other end was connected to a two-way value so that gas sample can be extracted. A 2-inch diameter hole was drilled to a desired depth using Geoprobe and the plastic tubing was inserted. The bottom 30 cm of the hole was filled with sand and the rest was filled with bentonite.

2.5. Water chemistry analysis

In the laboratory, the core samples for the pore water chemistry analysis were centrifuged to extract pore water. The water samples had been stored refrigerated until analysis. The centrifuged water yield varied and analyses were prioritized in the following order at GEUS: 1) Anions by an ion chromatography (IC); 2) ammonium by a Flow Injection Analysis (FIA); 3) Dissolved inorganic and organic carbon by an Infra-red (IR) detection on Shimadzu instrument; 4) Cations and trace elements by an Inductively Coupled Plasma - Mass Spectroscopy (ICP-MS); and 5) Water isotopes by a Cavity Ring Down Spectroscopy (CRDS).

2.6. Sediment chemistry analysis

A number of sediment chemistry parameters were analyzed for at GEUS:

1) Sediment Reduction Capacity was determined by oxidation with Ce(IV)

2) Fe(III) from less stable Fe-oxides, and Fe(II) compounds soluble at pH 3 were determined by extraction in formic acid at pH 3

3) Sediment inorganic and organic carbon as well as total sulfur was analyzed using a LECO furnace.

2.7. Nitrate reduction rate measurements and experiments

Rates of nitrate reduction were measured in the laboratory at GEUS using the acetylene-block method, where the transformation of N_2O to N_2 is blocked by adding acetylene to a closed vial containing sediment, 20 mL of a 1 mmol nitrate solution and a headspace, and following the production of N_2O for up to 2 weeks. The production of N_2O was followed by measuring the concentration in the known headspace of the vial and calculating the concentration in the known water volume.

Many of the incubations from LOOP2 show two phases in the N_2O development. An initial phase where the development in the N_2O concentrations is very slow, followed by a second phase where the rate in the N_2O development has shifted to a much higher level.

Incubations in other experiments, where the porewater was sampled over time, strongly indicate that the initial phase where a very slow rate in the N_2O development is seen is a phase where the last traces of O_2 in the incubation flask are removed. This is supported by the observation that this initial phase is sometimes not seen in incubations where high rates are measured, presumably because in these the traces of oxygen are rapidly removed by a reactive electron donor.

In the cases where there is a second rapid phase and it is adequately well described by a highly linear increase over at least three measuring points this second rapid phase has been used. This implies that many samples have not been included in the calculation of the final rates because the rate determined from only two points has a high probability of being underestimated.

The values are transformed from the rates of development in the N_2O concentration determined pr. gram of incubated sediment into µmolN_2O/l/yr assuming that there are 5.4 kg of sediment pr. liter of groundwater/porewater. In the final calculation this has been multiplied by 2, as each mole of N_2O represents 2 moles of NO_3^-

In addition to the rate measurements, experiments to address different aspects of the nitrate reduction and clarify the interpretation of the acetylene block N_2O rate measurements were carried out. The set of experiments were carried out on the mostly sandy sediment from LOOP6 and the mostly clayey sediment from LOOP4. The results are described and compared in the data report fro LOOP4.

Experiment 1: A 1 mM nitrate solution was added to the nitrate rate incubations, but to clarify whether the rate was dependent on the nitrate concentration we set up 12 sets of 2 incubations with different nitrate concentration (0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 1.0 (triplicates), 1.5, 2, 3, and 4 mM), using homogenized sediment from 2 core segment. In this experiment there was no acetylene added and the rates were derived from measuring the nitrate concentration in the water over time. To have enough water 300 g of sediment was added to 500 mL bottles and 130 mL of water was added.

Experiment 2: In experiment 2 the N_2/CO_2 atmosphere was changed to a pure He atmosphere and calcite was added as pH buffer instead of CO_2. In order to see if this change affected the rate, a set of incubations (triplicates) corresponding to the 1 mM nitrate incubations mentioned above, was set up using He as headspace gas and calcite as pH buffer.

To clarify to what extent the rate of nitrate reduction measured by the acetylene block method includes the part of the rate that could be associated with pyrite oxidation rather than oxidation of organic matter, Experiments 3 and 4 were set up in parallel. Experiment 3 in large bottles where the concentrations in the water could be followed to derive the rates and Experiment 4 where the acetylene block method was used.

Experiment 3: The experiment was done with sandy sediment from what based on the sediment color was most probably; oxidized, nitrate reducing and reduced sediment. For each type of sediment triplicates with 0, 125 and 250 mg of pyrite pr. 100 g of sediment were set up in 500 ml bottles. The pyrite added was crushed pyrite crystals that were sieved resulting in a powder with a BET surface area of 0.4 m^2/g. The headspace was He and 0.5 g of calcite powder pr. 100 g of sediment was added as pH buffer and no acetylene was added. The bottles were sampled for anions, Fe^{2+} and pH, one of the triplicates had an optode sticker mounted inside for measuring oxygen. The headspace gas was sampled and measured for N_2, O_2 and CO_2 by gas chromatography by TCD detection.

Experiment 4: This was parallel to experiment 3, with sediment from 3 apparent redox zones with varying amounts of pyrite, except done in the smaller 120 mL bottles with 100 g of sediment and here acetylene was added as in the normal rate measurements, but here the headspace was He and calcite was used as pH buffer.

2.8. Gas chemistry analysis

Gas samples were collected using plastic syringes. Before sampling, the gas wells were purged to remove the air in the gas well. The gas samples were collected in 3ml (for N_2O) and 6ml (for CO_2 and O_2) exetainers. The samples were duplicated.

The gas samples were analyzed for CO_2 and O_2 using gas chromatography (GC) with a TCD detector, N_2O was measured using a GC using cavity ringdown spectroscopy (CRDS) and N and O isotopes of N2O also using CRDS

2.9. Lithology description

Geological descriptions of the sediment samples were carried out at the laboratory at GEUS and shown in Appendix 1.

2.10. Redox probe and IP measurements

In February-March 2020, Ejlskov concluded that the redox-probe has been unstable since the summer of 2018 due to a hole in the insulation of the wire connecting the platinum electrode used for measurement of the signal. Ejlskov started troubleshooting the redox-probe and after a few rounds of amendments, the stability of the redox-probe was tested in LOOP2 in April 2020. More test and adjustments were carried out over the summer and autumn resulting in a dual-redox-probe. The final redox-probe measurements were carried out in February 2021 at sample points P1, P2, P2A, P3, P4, P5, P6, P7 and P8. At the field campaign in 2021 each sampling point were repeated twice and using the dual probe.

2.11. Water table

Water table measurements were carried out in the field directly in the boreholes left after each redox-probe measurement. At a few locations the level of the water table

was deeper than the bottom of the borehole. At other locations the borehole collapsed immediately after the drill stem was removed and it was not possible to measure the water table. The depth to the top of the collapse in the borehole was measured as it may be an indication of the water table. Water table measurements from the first round of redox probe measurements were used for the well panel plots.

To supplement and validate the information on the water table from field measurement an evaluation of the water table at each sampling point was carried out on existing data reported to the Jupiter database. The Jupiter borehole data including water level and a terrain grid were read into Geoscene 3D. Profiles crossing existing wells with water table levels and the MapField sampling points were drawn and at each sampling point the water table were interpreted taking the water level in adjacent well as well as the terrain level into account.

3. Results

3.1. Existing data

In total, water chemistry and groundwater table observations were available at 36 locations across the catchment (Figure1 and table1). Table 1 summarizes the availability of water chemistry and groundwater level data of LOOP6. Most LOOP monitoring stations have 1-4 filter depths for water sampling. Water chemistry was measured at 54 points and groundwater table is monitored at 31 points. The time-series of nitrate concentrations and groundwater table of each monitoring station are shown in Appendix 2 as an example.

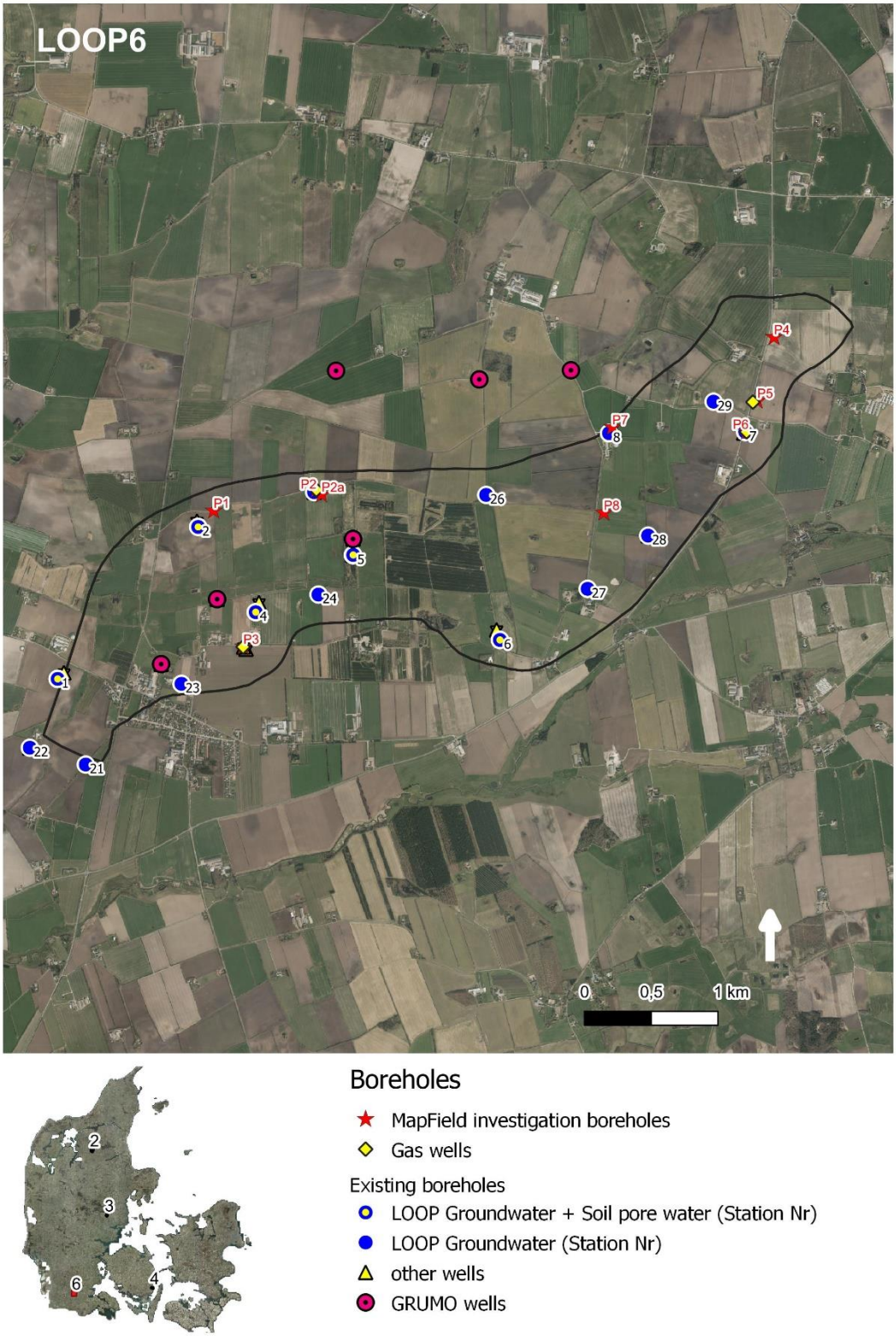


Figure 1. Overview existing boreholes with geochemical data, MapField investigation wells, and gas access well in LOOP6

Table 1. Summary of Existing data in LOOP6 (No = number)

LOOP Station No	Water chemistry							Groundwater table	
	DGU (X, Y)	Depth (m)	Data period	No of DO samples	No of NO_3^- samples	No of Fe (II) samples	No of SO_4^{2-} samples	DGU (X, Y)	(Nr of data) Data period
1	159. 926 (505829, 6101988)	2.2-2.5	1989-2012	7	58	42	32	159. 925 (505829, 6101988)	1989-2018 (1431)
	159. 927 (505829, 6101988)	1.2-1.5	1989-2019	98	166	142	45		
	159. 928 (505830, 6101989)	2.3-2.6	1989-1998	0	48	33	31		
	159. 929 (505830, 6101988)	1.3-1.6	1989-2019	97	166	141	46		
2	159. 931 (506882, 6103128)	2.0-2.3	1989-2012	5	56	42	33	159. 930 (506882, 6103127)	1989-2018 (1409)
	159. 932 (506882, 6103127)	1.0-1.3	1989-2019	100	164	140	42		
	159. 933 (506882, 6103127)	2.0-2.3	1989-1998	0	51	36	34		
	159. 934 (506882, 6103127)	1.0-1.3	1989-2019	93	148	125	35		
4	159. 941 (507313, 6102488)	3.2-3.5	1989-2019	57	159	139	47	159. 940 (507313, 6102488)	1989-2018 (1337)
	159. 942 (507313, 6102488)	2.2-2.5	1989-1998	0	31	18	16		
	159. 943 (507313, 6102488)	2.9-3.2	1989-2000	0	57	37	35		
	159. 944 (507313, 6102488)	1.9-2.2	1989-2000	0	30	17	16		
5	159. 946 (508040, 6102916)	2.1-2.4	1989-1998	0	46	35	32		
	159. 947 (508040, 6102916)	1.1-1.4	1989-2009	54	121	96	40		
	159. 948 (508043, 6102915)	2.1-2.4	1989-2000	0	46	35	33		
	159. 949 (508043, 6102917)	1.1-1.4	1989-2009	46	119	92	42		
6	159. 951 (509141, 6102280)	2.0-2.3	1989-1998	0	47	26	24	159. 950 (509141, 6102280)	1989-2018 (1350)
	159. 952 (509142, 6102280)	1.0-1.3	1989-2019	38	78	62	24		
	159. 953 (509141, 6102280)	2.0-2.3	1989-1998	0	35	15	15		
	159. 954 (509142, 6102280)	1.0-1.3	1989-1998	0	38	22	21		
7	159. 956 (510978, 6103839)	2.1-2.4	1989-2019	101	171	147	50		
	159. 957 (510979, 6103839)	1.1-1.4	1990-2000	0	22	12	11		
	159. 958 (510979, 6103839)	2.3-2.6	1989-2019	103	171	149	50		
	159. 959 (510979, 6103840)	1.3-1.6	1990-1998	0	24	14	13		
8	159. 961 (509959, 6103830)	2.3-2.6	1989-2019	103	166	145	46		
	159. 962 (509959, 6103830)	1.3-1.6	1989-1998	0	32	18	17		
	159. 963 (509959, 6103830)	2.2-2.5	1989-2019	102	166	145	47		
	159. 964 (509959, 6103830)	1.2-1.5	1989-1998	0	32	18	17		
21	159. 897 (506040, 6101346)	2.1-2.4	1989-2019	104	140	137	37	159. 899 (506040, 6101346)	1989-2018 (126)
	159. 898 (506040, 6101346)	1.1-1.4	1989-2000	0	30	23	21		
22	159. 900 (505620, 6101471)	2.5-2.8	1989-2019	103	140	134	37	159. 902 (505623, 6101472)	1989-2018 (123)
	159. 901 (505621, 6101472)	1.5-1.8	1989-1998	0	27	23	21		
23	159. 903 (506757, 6101951)	3.1-3.4	1989-1998	0	49	36	35	159. 905 (506757, 6101950)	1989-2018 (128)
	159. 904 (506757, 6101951)	2.1-2.4	1989-2019	84	117	111	29		
24	159. 907 (507783, 6102621)	3.9-4.2	1989-2019	98	130	129	35	159. 909 (507781, 6102620)	1989-2018 (124)
	159. 908 (507783, 6102620)	2.9-3.2	1989-1998	0	15	15	13		
25	159. 910 (507750, 6103387)	2.0-2.3	1989-2000	0	25	25	23	159. 912 (507750, 6103387)	1989-2018 (103)

	159. 911 (507750, 6103388)	1.0-1.3	1989-2019	101	141	131	34		
26	159. 913 (509041, 6103367)	2.1-2.4	1989-2019	99	136	130	34	159. 915 (509039, 6103367)	1990-2018 (123)
	159. 914 (509039, 6103367)	1.1-1.4	1989-2013	1	20	16	13		
27	159. 916 (509799, 6102663)	2.7-3.0	1989-2019	107	148	141	39	159. 918 (509799, 6102663)	1989-2018 (129)
	159. 917 (509799, 6102663)	1.7-2.0	1989-1998	0	19	15	13		
28	159. 919 (510252, 6103062)	2.4-2.7	1989-2000	0	32	26	24	159. 921 (510252, 6103062)	1989-2006 (72)
	159. 920 (510252, 6103062)	1.4-1.7	1990-2000	0	16	12	10		
29	159. 922 (510742, 6104066)	2.7-3.0	1989-1996	0	22	18	17	159. 924 (510748, 6104054)	1989-2009 (82)
GRUMO	159. 979 (506605, 6102099)	6.5-7.5	1989-2018	68	88	88	86	159. 979 (506605, 6102099)	1989-2018 (89)
	159. 982 (507024, 6102586)	29.0-30.0	1989-2017	94	152	152	123	159. 982 (507024, 6102586)	1989-2018 (157)
	159. 983 (508042, 6103037)	38.5-39.5	1989-2014	55	89	88	89	159. 983 (508042, 6103037)	1989-2009 (89)
	159. 984 (508042, 6103037)	2.2-3.2	1989-2018	73	119	119	102	159. 984 (508042, 6103037)	1989-2018 (126)
Others								159. 431 (510011, 6103090)	1977-1983 (34)
								159. 492 (506453, 6102802)	1976-2008 (283)
	159. 936 (507240, 6102212)	2.7-3.0	1989-2019	103	168	146	46	159. 935 (5072396, 6102212)	1989-2018 (1412)
								159. 1294 (507222, 6102228)	2014-2014 (53)
	159. 937 (507240, 6102212)	1.7-2.0	1989-1998	0	34	20	17		
	159. 938 (507240, 6102212)	2.4-2.7	1989-2019	101	166	144	47		
	159. 939 (507240, 6102212)	1.4-1.7	1989-2000	0	36	22	21		
	159. 978 (506605, 6102099)	54.0-55.0	1989-2017	63	114	114	98	159. 978 (506605, 6102099)	1989-2018 (136)
								159. 1292 (505876, 6102034)	2014-2014 (53)
								159. 1293 (506873, 6103170)	2014-2014 (53)
								159. 1295 (507337, 6102560)	2014-2014 (53)
								159. 945 (5080395, 6102916)	1990-2009 (982)
								159. 1297 (509117, 6102356)	2014-2014 (53)
								159. 955 (5109785, 6103840)	1989-2018 (1354)
								159. 1298 (510967, 6103832)	2014-2014 (53)
								159. 960 (5099585, 6103830)	1990-2018 (1404)
								159. 1299 (509977, 6103851)	2014-2014 (53)

3.2. Collected samples

In September 2019, the redox survey and core sampling were carried out at nine points (Figure 1 and Table 2). In total 63 core samples for pore water extraction, 31 for the nitrate reduction rate measurements, 46 samples for the sediment geochemistry and 54 samples for the lithology description were collected. In addition, for methodological evaluation of the N reduction rate measurement, 2 additional cores were collected at P2a and P6. In February 2021, the redox survey was carried out again in all locations. The data obtained in the redox probe survey of 2019 are unstable and should not be used in the interpretations and data analyses.

Table 2. Summary of core samples collected in the MapField project

Point	**DGU (X, Y)**	**Depth (m)**		**Number of samples**					**Total**
		Rx*	**Core****	**L[1)]**	**S[2)]**	**R[3)]**	**W[4)]**	**E[5)]**	
P1	159. 1692 (506998, 6103248)	10	6.1						
P2	159. 1693 (507767, 6103405)	15	4.88	5	4	3	5		**17**
P2a	159. 1694 (507806, 6103369)	15	4.88	4	3	4	4	1	**15**
P3	159. 1695 (507361, 6102278)	10	9.76	1	2	2	5		**10**
P4	159. 1696 (511195, 6104548)	7	7.32	9	8	8	11		**36**
P5	159. 1697 (511069, 6104072)	20	10.98	4	6		7		**17**
P6	159. 1698 (510989, 6103840)	10	8.54	10	9	4	10	1	**33**
P7	159. 1699 (509983, 6103873)	20	8.54	9	6	2	9		**26**
P8	159. 1700 (509920, 6103232)	20	7.32	9	3	3	6		**17**
total				**54**	**46**	**31**	**63**		**194**

[*] Redox probe, investigation depths measured in both surveys. The depths were the same for the two redox probe surveys; ** Core sampling; [1)] Lithology; [2)] Sediment chemistry; [3)] N reduction rate; [4)] Pore water chemistry; and [5)] N rate experiment

3.3. Overview of gas sample collection

During the Geoprobe campaign, four sets of gas sampling wells were installed at P2a, P3, P5 and P6 (Figure 1 and Table 3). A set of gas samples was collected a day after the installation. Two set of gas samples were collected in October 2019 and February 2020, respectively. It is important to note that the exetainers were not conditioned during these sampling. The exetainers were pre-vacuumed; however, a non-negligible amount of atmospheric air is often found to be remained in the exetainers, resulting in contamination of gas samples. The gas chemistry of the samples collected during 2019 and early 2020 should be carefully interpreted.

Table 3. Summary of gas access wells.

Location	Depth (m)	Nr of sample	Note
P2	0.5	4	
	1	2	
P3	0.5	5	
	1	3	
	1.5	1	Saturated
	2	3	
P5	0.5	5	
	1	5	
	2	5	
	3	5	
	4	3	
	5	1	Saturated
P6	0.5	3	
	1	3	
	1.5	1	Saturated
	2	1	Saturated

3.4. Interpretation of redox zones

The interpreted redox zones are shown in the chemistry tables in Appendix 2 where:

- Redox zone A is the oxic zone
- Redox zone B the anoxic nitrate reducing zone
- Redox zone C is the reduced zone

The evaluation of the redox zones is mainly based on the following indicators:

- The nitrate and sulfate, being redox sensitive water chemical compounds:
    - Stable high nitrate concentrations and low sulfate concentrations indicate oxic conditions

    - Low and/or deceasing nitrate concentrations over depth on agricultural fields indicate nitrate reducing conditions sometimes supported by increasing sulfate concentrations
    - Stable nitrate concentrations below 1-3 mg/l indicate reduced conditions as low concentrations of nitrate could be from ammonium oxidation during sampling
- The sediment content of Fe^{2+}/Fe_{total}, where even small amounts of extractable Fe(II) indicate lack of oxygen implying nitrate reducing or reduced conditions
- Color descriptions of the sediment, where reddish, orange, brown colors indicate oxic conditions, and olive, greyish colors indicate nitrate reducing or reduced conditions

The redox probe measurements seem to be able to detect the redox interfaces to some degree.

3.5. Overview of the MapField chemical analyses

The results from the analysis performed in 2019 on pore water and sediment samples are summarized in Appendix 3. Nitrate reduction rate, sediment chemistry and selected water chemistry data are displayed in Appendix 4 together with redox-probe measurement and IP measurements. The background shows the interpreted redox zones.

3.6. Results from nitrate reduction experiments

The results from the experiments are shown and discussed in the LOOP4 data report.

Appendices

Appendix 1: Lithological descriptions of the MapField samples

Appendix 2: Water chemistry of existing data (LOOP, GRUMO and others)

Excel file available

Appendix 3: Water (Appendix3-1 and 3-2), sediment chemistry (Appendix 3-3), denitrification rates (Appendix 3-4) collected in the MapField project

Appendix 4: Well panels illustrating all the collected parameters

The well panels display from left towards right:

Panel 1: Lithology (from Jupiter, Appendix 1)

Panel 2: Sediment color (from Jupiter, Appendix 1)

Panel 3: Redox probe measurement (unstable 2019 data, not to be used in the interpretations, and 2021 data use with caution)

Panel 4: Redox capacity (analysis pending) and N-rate (blue curve)

Panel 5: Sediment chemistry; extracted Fe(II) and total Fe as well as the Fe(II)/Fe(total).

Panel 6: Water isotopes; δO_{18} and δD.

Panel 7: Concentrations of ammonium.

Panel 8: Concentrations of Cl^-, NO_3^- and SO_4^{2-} and DOC

Panel 9: Concentrations of Ca^{2+}, Mg^{2+}, K^+ and Na^+

Appendix 1: Lithological descriptions of the MapField samples

BORERAPPORT

DGU arkivnr: 159. 1692

Borested : Bedstedvej 10
6240 Løgumkloster

Kommune : Tønder
Region : Syddanmark

Boringsdato : 24/9 2019

Boringsdybde : 6,1 meter

Terrænkote : 26,42 meter o. DNN

Brøndborer : Palle Ejlskov
MOB-nr :
BB-journr :
BB-bornr : LOOP6 - P1

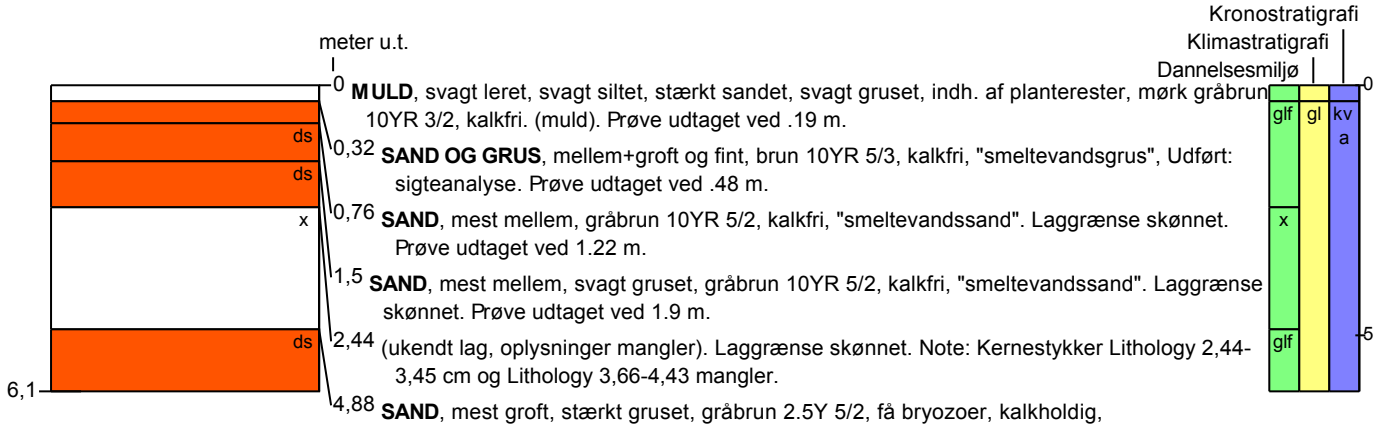
Prøver
- modtaget : 27/9 2019 **antal** : 5
- beskrevet : 11/11 2019 **af** : HJG
- antal gemt : 0

Formål : Undersøg./videnskab
Anvendelse :
Boremetode :

Kortblad : 1212IIISV
UTM-zone : 32
UTM-koord. : 506998, 6103248

Datum : EUREF89
Koordinatkilde : Brøndborer
Koordinatmetode : Landinspektør

Notater : Boringen består af kernestykker, hvor en del intervaller mangler. Sedimentet er ofte komprimeret (sjældnere ekspanderet) under prøvetagning. Begge dele giver usikkerhed om prøvedybder.



0 **MULD**, svagt leret, svagt siltet, stærkt sandet, svagt gruset, indh. af planterester, mørk gråbrun 10YR 3/2, kalkfri. (muld). Prøve udtaget ved .19 m.

0,32 **SAND OG GRUS**, mellem+groft og fint, brun 10YR 5/3, kalkfri, "smeltevandsgrus", Udført: sigteanalyse. Prøve udtaget ved .48 m.

0,76 **SAND**, mest mellem, gråbrun 10YR 5/2, kalkfri, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved 1.22 m.

1,5 **SAND**, mest mellem, svagt gruset, gråbrun 10YR 5/2, kalkfri, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved 1.9 m.

2,44 (ukendt lag, oplysninger mangler). Laggrænse skønnet. Note: Kernestykker Lithology 2,44-3,45 cm og Lithology 3,66-4,43 mangler.

4,88 **SAND**, mest groft, stærkt gruset, gråbrun 2.5Y 5/2, få bryozoer, kalkholdig, "smeltevandssand", Udført: sigteanalyse. Laggrænse skønnet. Prøve udtaget ved 5.91 m.

Aflejringsmiljø - Alder (klima-, krono-, litho-, biostratigrafi)

meter u.t.			
0	-	0,32	terrigen - postglacial - holocæn
0,32	-	2,44	glaciofluvial - glacial - kvartær
2,44	-	4,88	mangler - glacial - kvartær
4,88	-	6,1	glaciofluvial - glacial - kvartær

BORERAPPORT

DGU arkivnr: 159. 1693

Borested : Bedstedvej 10
6240 Løgumkloster

Kommune : Tønder
Region : Syddanmark

Boringsdato : 24/9 2019

Boringsdybde : 4,88 meter

Terrænkote : 27,84 meter o. DNN

Brøndborer : Palle Ejlskov
MOB-nr :
BB-journr :
BB-bornr : LOOP6 - P2

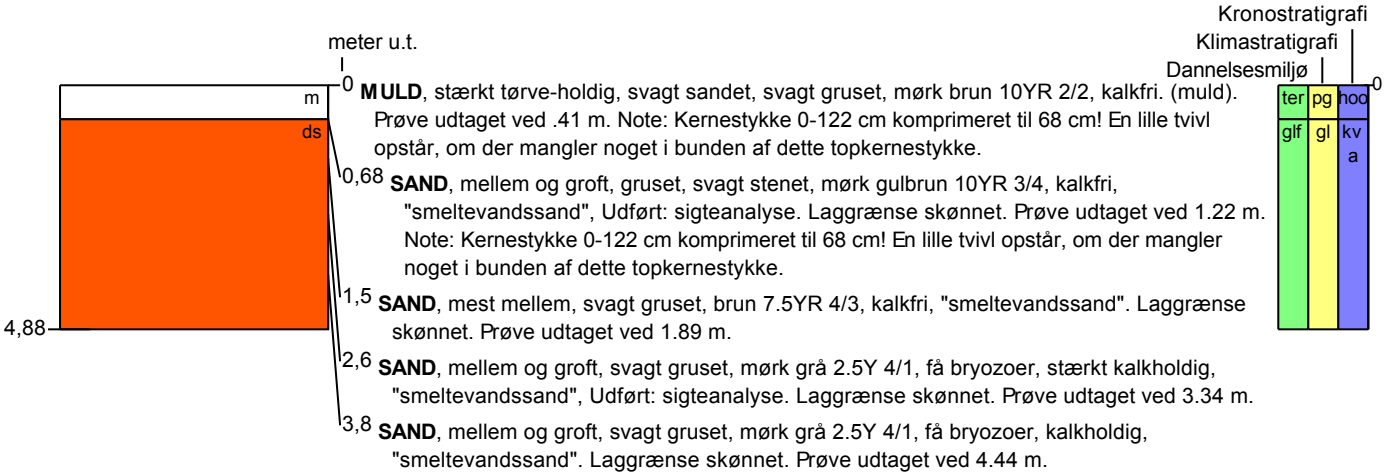
Prøver
- modtaget : 27/9 2019 **antal** : 5
- beskrevet : 11/11 2019 **af** : HJG
- antal gemt : 0

Formål : Undersøg./videnskab
Anvendelse :
Boremetode :

Kortblad : 1212IIISV
UTM-zone : 32
UTM-koord. : 507767, 6103405

Datum : EUREF89
Koordinatkilde : Brøndborer
Koordinatmetode : Landinspektør

Notater : Boringen består af kernestykker, hvor en del intervaller mangler. Sedimentet er ofte komprimeret (sjældnere ekspanderet) under prøvetagning. Begge dele giver usikkerhed om prøvedybder.



MULD, stærkt tørve-holdig, svagt sandet, svagt gruset, mørk brun 10YR 2/2, kalkfri. (muld). Prøve udtaget ved .41 m. Note: Kernestykke 0-122 cm komprimeret til 68 cm! En lille tvivl opstår, om der mangler noget i bunden af dette topkernestykke.

SAND, mellem og groft, gruset, svagt stenet, mørk gulbrun 10YR 3/4, kalkfri, "smeltevandssand", Udført: sigteanalyse. Laggrænse skønnet. Prøve udtaget ved 1.22 m. Note: Kernestykke 0-122 cm komprimeret til 68 cm! En lille tvivl opstår, om der mangler noget i bunden af dette topkernestykke.

SAND, mest mellem, svagt gruset, brun 7.5YR 4/3, kalkfri, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved 1.89 m.

SAND, mellem og groft, svagt gruset, mørk grå 2.5Y 4/1, få bryozoer, stærkt kalkholdig, "smeltevandssand", Udført: sigteanalyse. Laggrænse skønnet. Prøve udtaget ved 3.34 m.

SAND, mellem og groft, svagt gruset, mørk grå 2.5Y 4/1, få bryozoer, kalkholdig, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved 4.44 m.

Aflejringsmiljø - Alder (klima-, krono-, litho-, biostratigrafi)

meter u.t.

0 - 0,68 terrigen - postglacial - holocæn
0,68 - 4,88 glaciofluvial - glacial - kvartær

BORERAPPORT

DGU arkivnr: 159. 1694

Borested : Bedstedvej 10
6240 Løgumkloster

Kommune : Tønder
Region : Syddanmark

Boringsdato : 24/9 2019
Boringsdybde : 4,88 meter
Terrænkote : 27,5 meter o. DNN

Brøndborer : Palle Ejlskov
MOB-nr :
BB-journr :
BB-bornr : LOOP6 - P2a

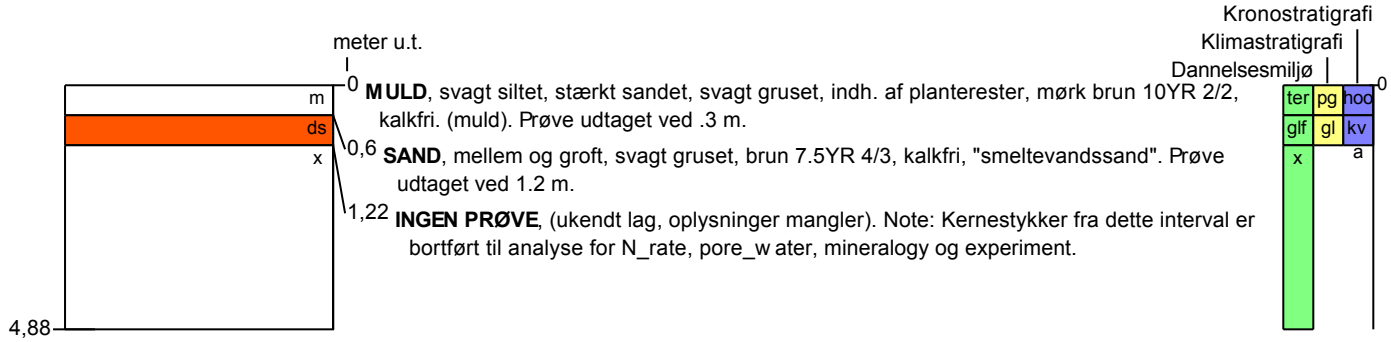
Prøver
- modtaget : 27/9 2019 **antal** : 2
- beskrevet : 11/11 2019 **af** : HJG
- antal gemt : 0

Formål : Undersøg./videnskab
Anvendelse :
Boremetode :

Kortblad : 1212IIISV
UTM-zone : 32
UTM-koord. : 507806, 6103369

Datum : EUREF89
Koordinatkilde : Brøndborer
Koordinatmetode : Landinspektør

Notater : Boringen består af 1 kernestykke 122 cm langt. Den resterende del af boringen er bortført til analyser for N_rate, pore_water, mineralogy og experiment.



MULD, svagt siltet, stærkt sandet, svagt gruset, indh. af planterester, mørk brun 10YR 2/2, kalkfri. (muld). Prøve udtaget ved .3 m.

SAND, mellem og groft, svagt gruset, brun 7.5YR 4/3, kalkfri, "smeltevandssand". Prøve udtaget ved 1.2 m.

INGEN PRØVE, (ukendt lag, oplysninger mangler). Note: Kernestykker fra dette interval er bortført til analyse for N_rate, pore_w ater, mineralogy og experiment.

Aflejringsmiljø - Alder (klima-, krono-, litho-, biostratigrafi)

meter u.t.			
0	-	0,6	terrigen - postglacial - holocæn
0,6	-	1,22	glaciofluvial - glacial - kvartær
1,22	-	4,88	mangler

BORERAPPORT

DGU arkivnr: 159. 1695

Borested : Bjergvej 17
6240 Løgumkloster

Kommune : Tønder
Region : Syddanmark

Boringsdato : 24/9 2019
Boringsdybde : 9,76 meter
Terrænkote : 27,3 meter o. DNN

Brøndborer : Palle Ejlskov
MOB-nr :
BB-journr :
BB-bornr : LOOP6 - P3

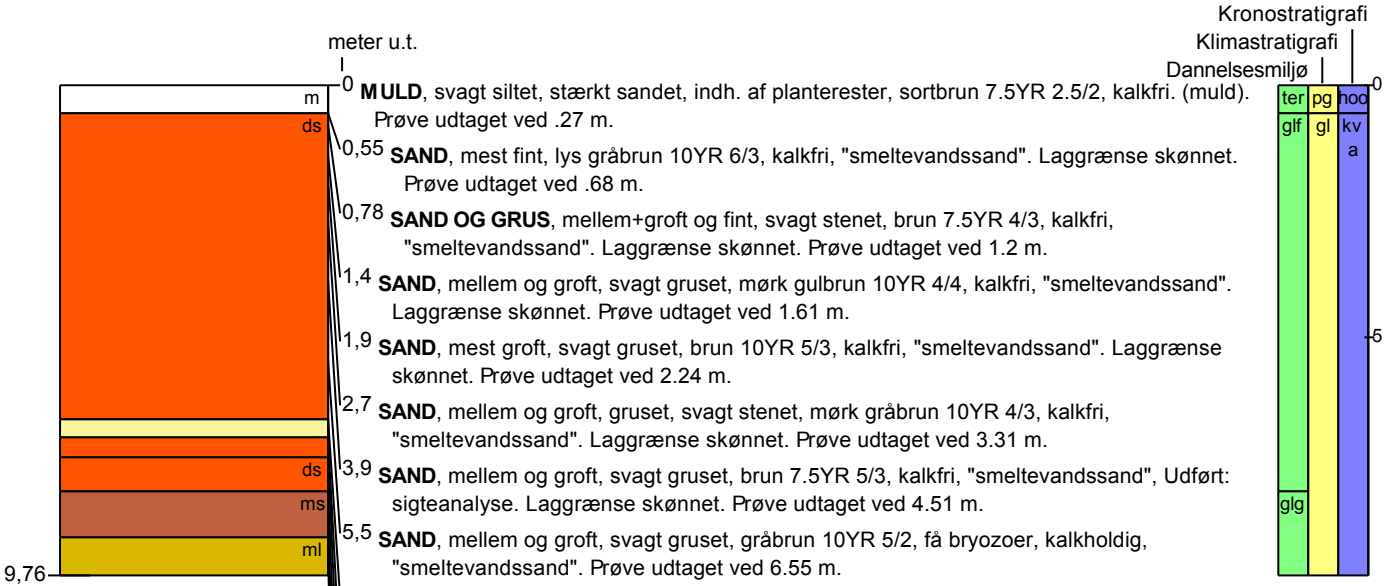
Prøver
- modtaget : 27/9 2019 **antal** : 15
- beskrevet : 11/11 2019 **af** : HJG
- antal gemt : 0

Formål : Undersøg./videnskab
Anvendelse :
Boremetode :

Kortblad : 1212IIISV
UTM-zone : 32
UTM-koord. : 507361, 6102278

Datum : EUREF89
Koordinatkilde : Brøndborer
Koordinatmetode : Landinspektør

Notater : Boringen består af kernestykker, hvor en del intervaller mangler. Sedimentet er ofte komprimeret (sjældnere ekspanderet) under prøvetagning. Begge dele giver usikkerhed om prøvedybder.



0 **MULD**, svagt siltet, stærkt sandet, indh. af planterester, sortbrun 7.5YR 2.5/2, kalkfri. (muld). Prøve udtaget ved .27 m.

0,55 **SAND**, mest fint, lys gråbrun 10YR 6/3, kalkfri, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved .68 m.

0,78 **SAND OG GRUS**, mellem+groft og fint, svagt stenet, brun 7.5YR 4/3, kalkfri, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved 1.2 m.

1,4 **SAND**, mellem og groft, svagt gruset, mørk gulbrun 10YR 4/4, kalkfri, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved 1.61 m.

1,9 **SAND**, mest groft, svagt gruset, brun 10YR 5/3, kalkfri, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved 2.24 m.

2,7 **SAND**, mellem og groft, gruset, svagt stenet, mørk gråbrun 10YR 4/3, kalkfri, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved 3.31 m.

3,9 **SAND**, mellem og groft, svagt gruset, brun 7.5YR 5/3, kalkfri, "smeltevandssand", Udført: sigteanalyse. Laggrænse skønnet. Prøve udtaget ved 4.51 m.

5,5 **SAND**, mellem og groft, svagt gruset, gråbrun 10YR 5/2, få bryozoer, kalkholdig, "smeltevandssand". Prøve udtaget ved 6.55 m.

6,65 **SILT**, leret, svagt sandet, brun 10YR 5/3, kalkfri, "smeltevandssilt". Laggrænse skønnet. Prøve udtaget ved 6.68 m.

7 **SAND OG GRUS**, groft og fint, svagt sandet, brun 7.5YR 4/3, kalkfri, "smeltevandsgrus". Laggrænse skønnet. Prøve udtaget ved 7.32 m.

7,4 **SAND**, mest mellem, sort 10YR 2/1, kalkfri, "smeltevandssand", Udført: sigteanalyse. Laggrænse skønnet. Prøve udtaget ved 7.66 m.

8,1 **SAND**, leret, siltet, svagt gruset, grå 5Y 5/1, stærkt kalkholdig, "morænesand". Laggrænse skønnet. Prøve udtaget ved 8.68 m.

8,8 **SAND**, svagt leret, siltet, svagt gruset, grå 5Y 5/1, stærkt kalkholdig, "morænesand". Laggrænse skønnet. Prøve udtaget ved 8.93 m.

9 **LER**, siltet, stærkt sandet, svagt gruset, mørk grå 5Y 4/1, stærkt kalkholdig, "moræneler". Laggrænse skønnet. Prøve udtaget ved 9.32 m.

9,5 **LER**, siltet, stærkt sandet, svagt gruset, slirer af sand, mørk grå 5Y 4/1, stærkt kalkholdig, "moræneler". Laggrænse skønnet. Prøve udtaget ved 9.67 m.

Aflejringsmiljø - Alder (klima-, krono-, litho-, biostratigrafi)

meter u.t.		
0 - 0,55	terrigen - postglacial - holocæn	
0,55 - 8,1	glaciofluvial - glacial - kvartær	
8,1 - 9,76	glacigen - glacial - kvartær	

BORERAPPORT

DGU arkivnr: 159. 1696

Borested : Rangstrupvej 4
6230 Rødekro

Kommune : Aabenraa
Region : Syddanmark

Boringsdato : 25/9 2019
Boringsdybde : 7,32 meter
Terrænkote : 45,03 meter o. DNN

Brøndborer : Palle Ejlskov
MOB-nr :
BB-journr :
BB-bornr : LOOP6 - P4

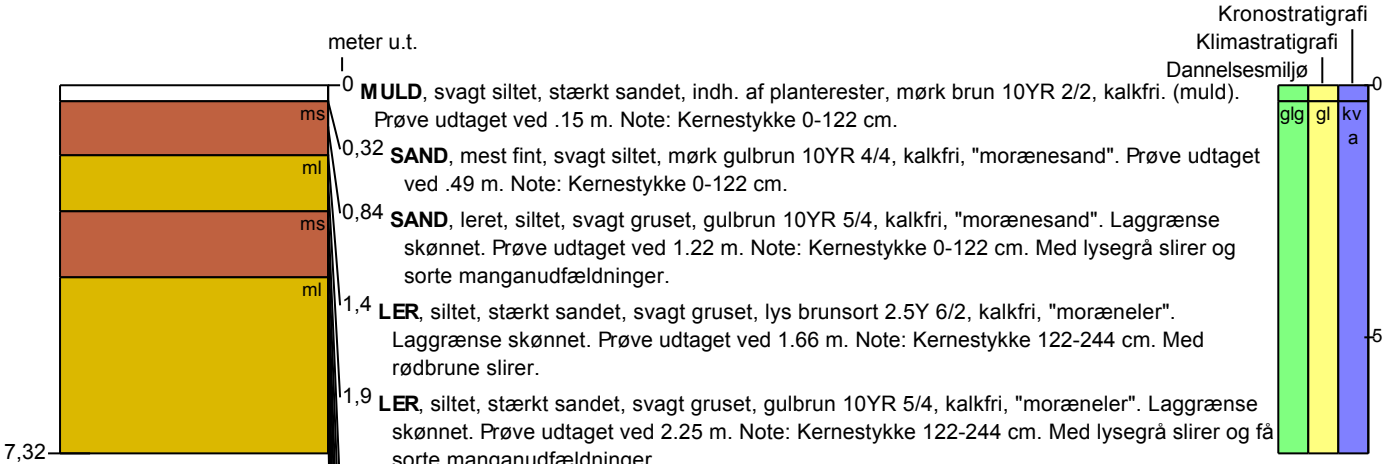
Prøver
- modtaget : 27/9 2019 **antal** : 14
- beskrevet : 11/11 2019 **af** : HJG
- antal gemt : 0

Formål : Undersøg./videnskab
Anvendelse :
Boremetode :

Kortblad : 1212IIISV
UTM-zone : 32
UTM-koord. : 511195, 6104548

Datum : EUREF89
Koordinatkilde : Brøndborer
Koordinatmetode : Landinspektør

Notater : Boringen består af kernestykker, hvor en del intervaller mangler. Sedimentet er ofte komprimeret (sjældnere ekspanderet) under prøvetagning. Begge dele giver usikkerhed om prøvedybder.



0 **MULD**, svagt siltet, stærkt sandet, indh. af planterester, mørk brun 10YR 2/2, kalkfri. (muld). Prøve udtaget ved .15 m. Note: Kernestykke 0-122 cm.

0,32 **SAND**, mest fint, svagt siltet, mørk gulbrun 10YR 4/4, kalkfri, "morænesand". Prøve udtaget ved .49 m. Note: Kernestykke 0-122 cm.

0,84 **SAND**, leret, siltet, svagt gruset, gulbrun 10YR 5/4, kalkfri, "morænesand". Laggrænse skønnet. Prøve udtaget ved 1.22 m. Note: Kernestykke 0-122 cm. Med lysegrå slirer og sorte manganudfældninger.

1,4 **LER**, siltet, stærkt sandet, svagt gruset, lys brunsort 2.5Y 6/2, kalkfri, "moræneler". Laggrænse skønnet. Prøve udtaget ved 1.66 m. Note: Kernestykke 122-244 cm. Med rødbrune slirer.

1,9 **LER**, siltet, stærkt sandet, svagt gruset, gulbrun 10YR 5/4, kalkfri, "moræneler". Laggrænse skønnet. Prøve udtaget ved 2.25 m. Note: Kernestykke 122-244 cm. Med lysegrå slirer og få sorte manganudfældninger.

2,5 **SAND**, stærkt leret, siltet, svagt gruset, mørk gulbrun 10YR 4/4, kalkfri, "morænesand". Laggrænse skønnet. Prøve udtaget ved 2.7 m. Note: Kernestykke 244-366 cm.

3,1 **SAND**, stærkt leret, siltet, svagt gruset, gulbrun 10YR 5/4, kalkfri, "morænesand". Laggrænse skønnet. Prøve udtaget ved 3.47 m. Note: Kernestykke 244-366 cm.

3,8 **LER**, siltet, stærkt sandet, svagt gruset, brun 10YR 5/3, kalkfri, "moræneler". Laggrænse skønnet. Prøve udtaget ved 4.02 m. Note: Kernestykke 366-488 cm.

4,3 **LER**, siltet, stærkt sandet, svagt gruset, brun 10YR 5/3, kalkfri, "moræneler". Laggrænse skønnet. Prøve udtaget ved 4.69 m. Note: Kernestykke 366-488 cm.

5 **LER**, siltet, stærkt sandet, svagt gruset, brun 10YR 5/3, kalkfri, "moræneler". Laggrænse skønnet. Prøve udtaget ved 5.37 m. Note: Kernestykke 488-610 cm.

5,6 **LER**, siltet, stærkt sandet, svagt gruset, gulbrun 10YR 5/4, kalkfri, "moræneler". Laggrænse skønnet. Prøve udtaget ved 5.91 m. Note: Kernestykke 488-610 cm.

6 **LER**, siltet, stærkt sandet, svagt gruset, mørk grå 5Y 4/1, kalkfri, "moræneler". Laggrænse skønnet. Prøve udtaget ved 6.04 m. Note: Kernestykke 610-732 cm. Med blåt påskrevet 5,84-6,04 ??? ekspanderet.

6,2 **LER**, siltet, stærkt sandet, svagt gruset, mørk grå 5Y 4/1, kalkfri, "moræneler". Laggrænse skønnet. Prøve udtaget ved 6.45 m. Note: Kernestykke 610-732 cm.

6,7 **LER**, siltet, stærkt sandet, svagt gruset, mørk grå 5Y 4/1, kalkfri, "moræneler". Laggrænse skønnet. Prøve udtaget ved 7.13 m. Note: Kernestykke 610-732 cm.

Aflejringsmiljø - Alder (klima-, krono-, litho-, biostratigrafi)

meter u.t.			
0	-	0,32	terrigen - postglacial - holocæn
0,32	-	7,32	glacigen - glacial - kvartær

BORERAPPORT

DGU arkivnr: 159. 1697

Borested : Ørslewej 11
6230 Rødekro

Kommune : Aabenraa
Region : Syddanmark

Boringsdato : 25/9 2019

Boringsdybde : 10,98 meter

Terrænkote : 41,12 meter o. DNN

Brøndborer : Palle Ejlskov
MOB-nr :
BB-journr :
BB-bornr : LOOP6 - P5

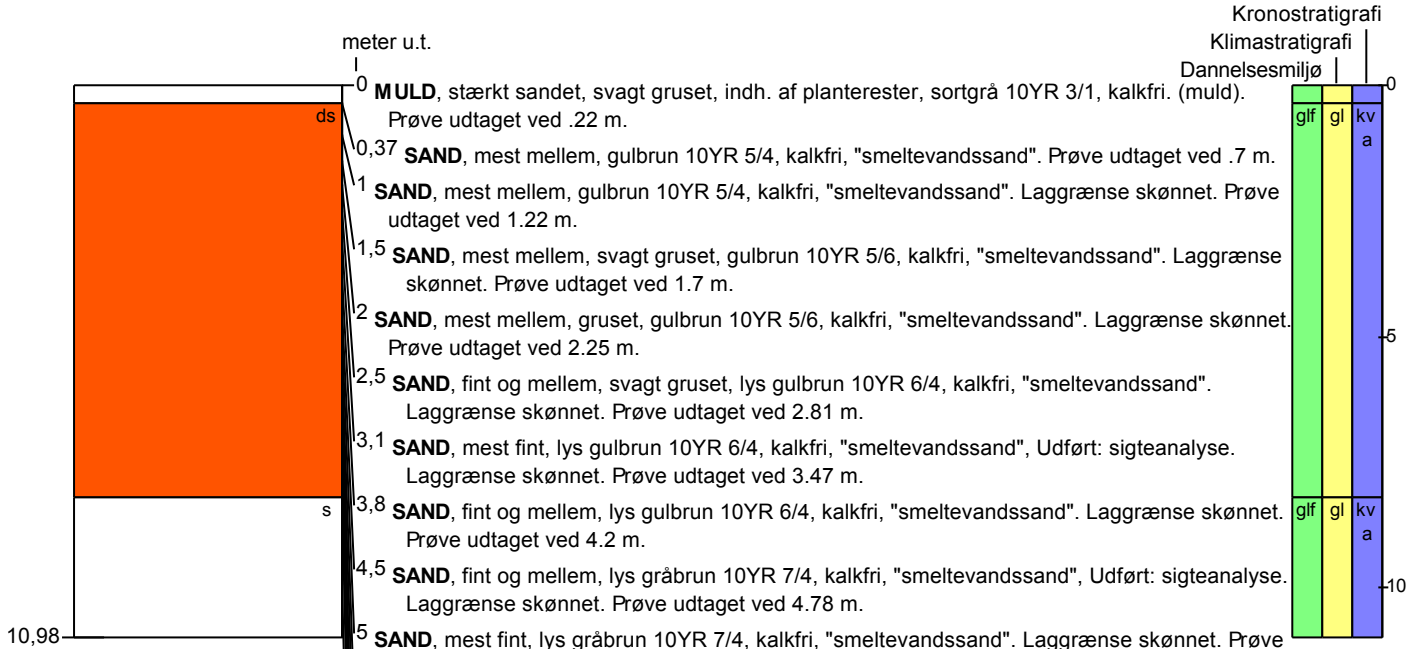
Prøver
- modtaget : 27/9 2019 **antal** : 17
- beskrevet : 11/11 2019 **af** : HJG
- antal gemt : 0

Formål : Undersøg./videnskab
Anvendelse :
Boremetode :

Kortblad : 1212IIISV
UTM-zone : 32
UTM-koord. : 511069, 6104072

Datum : EUREF89
Koordinatkilde : Brøndborer
Koordinatmetode : Landinspektør

Notater : Boringen består af kernestykker, hvor en del intervaller mangler. Sedimentet er ofte komprimeret (sjældnere ekspanderet) under prøvetagning. Begge dele giver usikkerhed om prøvedybder.



0 **MULD**, stærkt sandet, svagt gruset, indh. af planterester, sortgrå 10YR 3/1, kalkfri. (muld). Prøve udtaget ved .22 m.

0,37 **SAND**, mest mellem, gulbrun 10YR 5/4, kalkfri, "smeltevandssand". Prøve udtaget ved .7 m.

1 **SAND**, mest mellem, gulbrun 10YR 5/4, kalkfri, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved 1.22 m.

1,5 **SAND**, mest mellem, svagt gruset, gulbrun 10YR 5/6, kalkfri, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved 1.7 m.

2 **SAND**, mest mellem, gruset, gulbrun 10YR 5/6, kalkfri, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved 2.25 m.

2,5 **SAND**, fint og mellem, svagt gruset, lys gulbrun 10YR 6/4, kalkfri, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved 2.81 m.

3,1 **SAND**, mest fint, lys gulbrun 10YR 6/4, kalkfri, "smeltevandssand", Udført: sigteanalyse. Laggrænse skønnet. Prøve udtaget ved 3.47 m.

3,8 **SAND**, fint og mellem, lys gulbrun 10YR 6/4, kalkfri, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved 4.2 m.

4,5 **SAND**, fint og mellem, lys gråbrun 10YR 7/4, kalkfri, "smeltevandssand", Udført: sigteanalyse. Laggrænse skønnet. Prøve udtaget ved 4.78 m.

5 **SAND**, mest fint, lys gråbrun 10YR 7/4, kalkfri, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved 5.19 m.

5,6 **SAND**, fint og mellem, lys gråbrun 10YR 7/4, kalkfri, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved 5.91 m.

6,1 **SAND**, fint og mellem, lys gråbrun 10YR 6/3, kalkfri, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved 6.38 m.

6,6 **SAND**, mest mellem, rødbrun 5YR 4/4, kalkfri, "smeltevandssand", Udført: sigteanalyse. Laggrænse skønnet. Prøve udtaget ved 6.78 m.

7 **SAND**, mest mellem, svagt gruset, mørk gulbrun 10YR 4/6, kalkfri, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved 7.2 m.

7,3 **SAND**, mest mellem, lys brungrå 10YR 7/2, kalkfri, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved 7.47 m.

8,2 **SAND**, mest mellem, grå 2.5Y 6/1, stærkt kvarts-holdigt, svagt glimmer-holdigt, kalkfri. (sand). Laggrænse skønnet. Prøve udtaget ved 8.91 m.

9,9 **SAND**, mest fint, grå 2.5Y 6/1, stærkt kvarts-holdigt, svagt glimmer-holdigt, kalkfri. (sand), Udført: sigteanalyse. Laggrænse skønnet. Prøve udtaget ved 10.89 m. Note: Udelukkende kvarts, velsorteret og afrundede korn. Miocæn? Prækvartærkote -75- -100.

fortsættes..

BORERAPPORT

DGU arkivnr: 159. 1697

Aflejringsmiljø - Alder **(klima-, krono-, litho-, biostratigrafi)**

meter u.t.			
0	-	0,37	terrigen - postglacial - holocæn
0,37	-	8,2	glaciofluvial - glacial - kvartær
8,2	-	10,98	ant. glaciofluvial - ant. glacial - ant. kvartær

BORERAPPORT

DGU arkivnr: 159. 1698

Borested : Ørslewej 11
6230 Rødekro

Kommune : Aabenraa
Region : Syddanmark

Boringsdato : 25/9 2019

Boringsdybde : 8,54 meter

Terrænkote : 32,21 meter o. DNN

Brøndborer : Palle Ejlskov
MOB-nr :
BB-journr :
BB-bornr : LOOP6 - P6

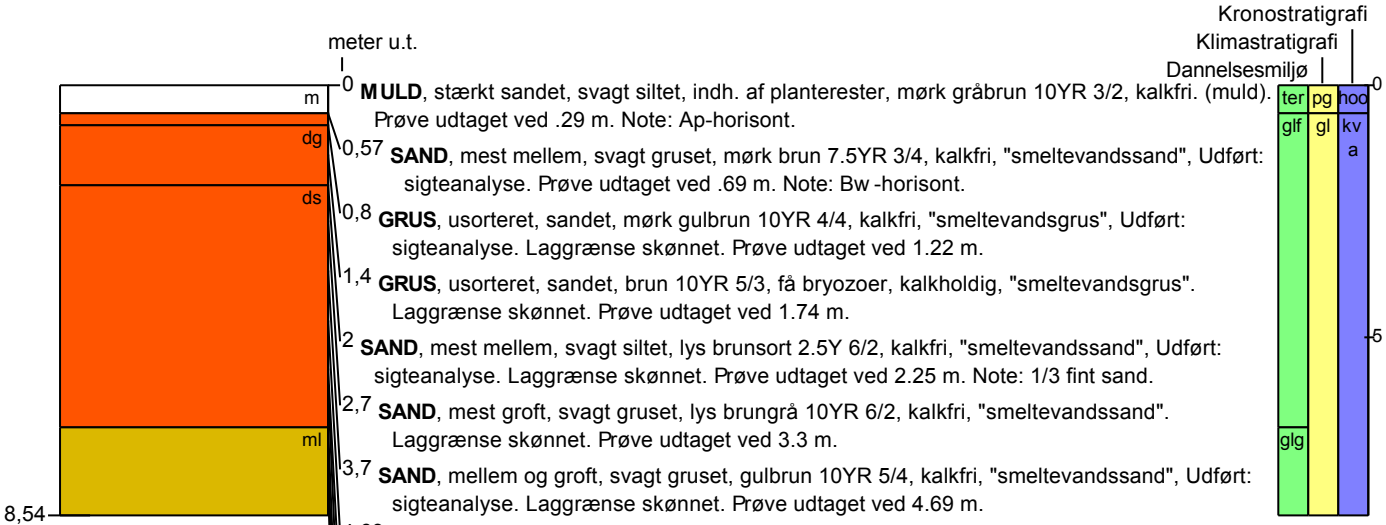
Prøver
- modtaget : 27/9 2019 **antal** : 13
- beskrevet : 11/11 2019 **af** : HJG
- antal gemt : 0

Formål : Undersøg./videnskab
Anvendelse :
Boremetode :

Kortblad : 1212IIISV
UTM-zone : 32
UTM-koord. : 510989, 6103840

Datum : EUREF89
Koordinatkilde : Brøndborer
Koordinatmetode : Landinspektør

Notater : Boringen består af kernestykker, hvor en del intervaller mangler. Sedimentet er ofte komprimeret (sjældnere ekspanderet) under prøvetagning. Begge dele giver usikkerhed om prøvedybder.



0 **MULD**, stærkt sandet, svagt siltet, indh. af planterester, mørk gråbrun 10YR 3/2, kalkfri. (muld). Prøve udtaget ved .29 m. Note: Ap-horisont.

0,57 **SAND**, mest mellem, svagt gruset, mørk brun 7.5YR 3/4, kalkfri, "smeltevandssand", Udført: sigteanalyse. Prøve udtaget ved .69 m. Note: Bw -horisont.

0,8 **GRUS**, usorteret, sandet, mørk gulbrun 10YR 4/4, kalkfri, "smeltevandsgrus", Udført: sigteanalyse. Laggrænse skønnet. Prøve udtaget ved 1.22 m.

1,4 **GRUS**, usorteret, sandet, brun 10YR 5/3, få bryozoer, kalkholdig, "smeltevandsgrus". Laggrænse skønnet. Prøve udtaget ved 1.74 m.

2 **SAND**, mest mellem, svagt siltet, lys brunsort 2.5Y 6/2, kalkfri, "smeltevandssand", Udført: sigteanalyse. Laggrænse skønnet. Prøve udtaget ved 2.25 m. Note: 1/3 fint sand.

2,7 **SAND**, mest groft, svagt gruset, lys brungrå 10YR 6/2, kalkfri, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved 3.3 m.

3,7 **SAND**, mellem og groft, svagt gruset, gulbrun 10YR 5/4, kalkfri, "smeltevandssand", Udført: sigteanalyse. Laggrænse skønnet. Prøve udtaget ved 4.69 m.

4,69 **SAND**, mest mellem, gråbrun 10YR 5/2, få bryozoer, stærkt kalkholdig, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved 5.61 m.

5,63 **SAND**, mellem og groft, stærkt gruset, mørk gråbrun 10YR 3/3, få bryozoer, kalkholdig, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved 5.92 m.

5,97 **SAND**, mellem og groft, stærkt gruset, grå 2.5Y 5/1, kalkholdig, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved 6.45 m.

6,8 **LER**, siltet, sandet, svagt gruset, mørk grå 2.5Y 4/1, stærkt kalkholdig, "moræneler". Laggrænse skønnet. Prøve udtaget ved 7.13 m.

7,4 **LER**, siltet, sandet, svagt gruset, mørk grå 2.5Y 4/1, svagt indh.af kalkklaster, stærkt kalkholdig, "moræneler". Laggrænse skønnet. Prøve udtaget ved 7.7 m.

8 **LER**, siltet, sandet, svagt gruset, mørk grå 2.5Y 4/1, svagt indh.af kalkklaster, stærkt kalkholdig, "moræneler". Laggrænse skønnet. Prøve udtaget ved 8.45 m.

Aflejringsmiljø - Alder (klima-, krono-, litho-, biostratigrafi)

meter u.t.			
0	-	0,57	terrigen - postglacial - holocæn
0,57	-	6,8	glaciofluvial - glacial - kvartær
6,8	-	8,54	glacigen - glacial - kvartær

BORERAPPORT

DGU arkivnr: 159. 1699

Borested : Hinderupvej 2
6230 Rødekro

Kommune : Aabenraa
Region : Syddanmark

Boringsdato : 26/9 2019
Boringsdybde : 8,54 meter
Terrænkote : 31,27 meter o. DNN

Brøndborer : Palle Ejlskov
MOB-nr :
BB-journr :
BB-bornr : LOOP6 - P7

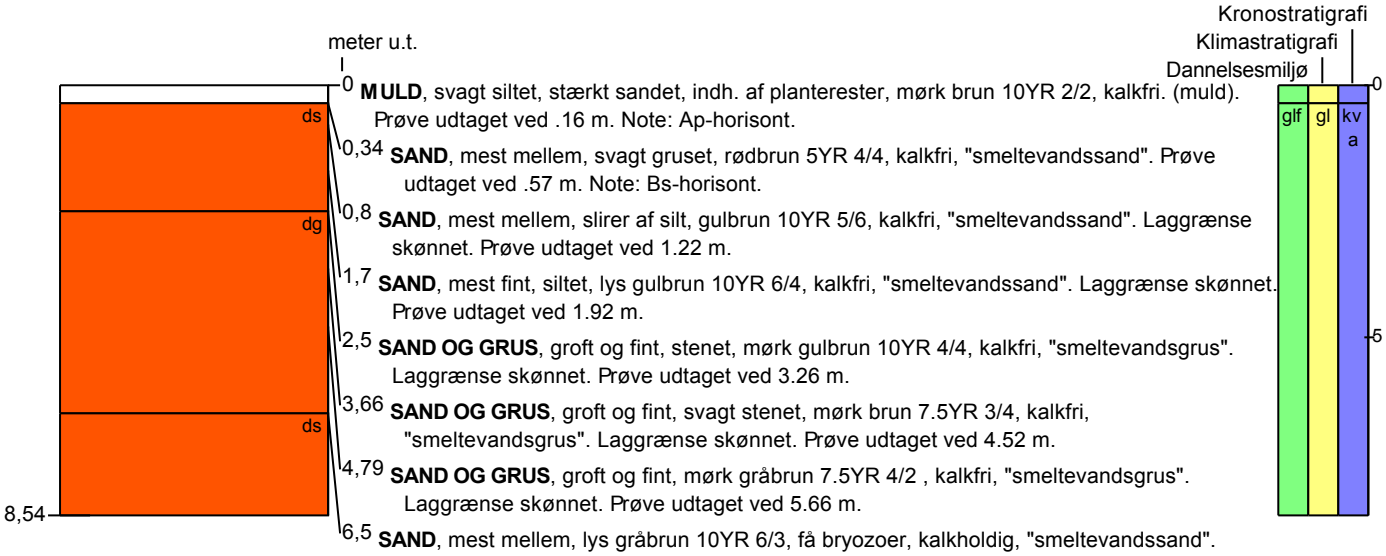
Prøver
- modtaget : 27/9 2019 **antal** : 8
- beskrevet : 8/11 2019 **af** : HJG
- antal gemt : 0

Formål : Undersøg./videnskab
Anvendelse :
Boremetode :

Kortblad : 1212IIISV
UTM-zone : 32
UTM-koord. : 509983, 6103873

Datum : EUREF89
Koordinatkilde : Brøndborer
Koordinatmetode : Landinspektør

Notater : Boringen består af kernestykker, hvor en del intervaller mangler. Sedimentet er ofte komprimeret (sjældnere ekspanderet) under prøvetagning. Begge dele giver usikkerhed om dybder.



0 **MULD**, svagt siltet, stærkt sandet, indh. af planterester, mørk brun 10YR 2/2, kalkfri. (muld). Prøve udtaget ved .16 m. Note: Ap-horisont.

0,34 **SAND**, mest mellem, svagt gruset, rødbrun 5YR 4/4, kalkfri, "smeltevandssand". Prøve udtaget ved .57 m. Note: Bs-horisont.

0,8 **SAND**, mest mellem, slirer af silt, gulbrun 10YR 5/6, kalkfri, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved 1.22 m.

1,7 **SAND**, mest fint, siltet, lys gulbrun 10YR 6/4, kalkfri, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved 1.92 m.

2,5 **SAND OG GRUS**, groft og fint, stenet, mørk gulbrun 10YR 4/4, kalkfri, "smeltevandsgrus". Laggrænse skønnet. Prøve udtaget ved 3.26 m.

3,66 **SAND OG GRUS**, groft og fint, svagt stenet, mørk brun 7.5YR 3/4, kalkfri, "smeltevandsgrus". Laggrænse skønnet. Prøve udtaget ved 4.52 m.

4,79 **SAND OG GRUS**, groft og fint, mørk gråbrun 7.5YR 4/2 , kalkfri, "smeltevandsgrus". Laggrænse skønnet. Prøve udtaget ved 5.66 m.

6,5 **SAND**, mest mellem, lys gråbrun 10YR 6/3, få bryozoer, kalkholdig, "smeltevandssand". Laggrænse skønnet. Prøve udtaget ved 7.17 m.

Aflejringsmiljø - Alder (klima-, krono-, litho-, biostratigrafi)

meter u.t.			
0	-	0,34	terrigen - postglacial - holocæn
0,34	-	8,54	glaciofluvial - glacial - kvartær

BORERAPPORT

DGU arkivnr: 159. 1670

Borested : Bylderup-Bov, 6372

Ravsted Hovedgade 68, Ravsted

Kommune : Aabenraa
Region : Syddanmark

Boringsdato :

Boringsdybde :

Terrænkote : 22,8 meter o. DNN

Brøndborer : Brøndboreren er ukendt
MOB-nr :
BB-journr :
BB-bornr :

Prøver
- modtaget :
- beskrevet :
- antal gemt :

Formål : Forurening/miljø
Anvendelse :
Boremetode :

Kortblad : 1212IIISV
UTM-zone : 32
UTM-koord. : 507876, 6095570

Datum : EUREF89
Koordinatkilde :
Koordinatmetode : Centerkoordinat

		Ro-vandstand	Pejledato	Ydelse	Sænkning	Pumpetid
Indtag 1	(seneste)	2,13 meter u.t.	28/3 2003			

ø ukendt

meter u.t.
I

Appendix 3: Water (Appendix3-1 and 3-2), sediment chemistry (Appendix 3-3), denitrification rates (Appendix 3-4) collected in the MapField project

Appendix3-1. Results of anions, dissolved organic carbon (DOC), dissolved inorganic carbon (DIC), stable isotope of water of LOOP6

LOOPNr	ID	method	DGUnr	Depth (m)	mg/L						µg/L	mg/L		‰	
					F^-	Cl^-	Br^-	NO_3^-	PO_4^-	SO_4^{2-}	NH_4^+	DOC	TIC	δO18	δD
LOOP6	P1	GeoProbe	159. 1692	1.95	0.30	22.65	-0.05	45.96	0.02	29.75	99.14	34.38	40.68	-6.36	-41.65
LOOP6	P1	GeoProbe	159. 1692	1.945	0.30	22.65	<0,05	45.96	0.02	29.75	99.14	34.38	40.68	-6.36	-41.65
LOOP6	P1	GeoProbe	159. 1692	2.41	0.71	41.38	0.24	0.20	0.02	144.42	521.82				
LOOP6	P1	GeoProbe	159. 1692	2.405	0.71	41.38	0.24	0.20	0.02	144.42	521.82				
LOOP6	P1	GeoProbe	159. 1692	3.62	0.72	28.11	0.06	0.09	0.02	89.24	711.53	63.86	17.08	-6.69	-45.37
LOOP6	P1	GeoProbe	159. 1692	3.615	0.72	28.11	0.06	0.09	0.02	89.24	711.53	63.86	17.08	-6.69	-45.37
LOOP6	P1	GeoProbe	159. 1692	4.84	1.54	34.47	0.17	0.29	0.02	89.49	539.27	74.59	36.79	-6.92	-46.56
LOOP6	P1	GeoProbe	159. 1692	4.835	1.54	34.47	0.17	0.29	0.02	89.49	539.27	74.59	36.79	-6.92	-46.56
LOOP6	P1	GeoProbe	159. 1692	6.06	0.69	22.00	0.07	0.05	0.02	65.58	193.82	41.92	34.83	-6.87	-46.34
LOOP6	P1	GeoProbe	159. 1692	6.055	0.69	22.00	0.07	0.05	0.02	65.58	193.82	41.92	34.83	-6.87	-46.34
LOOP6	P2	GeoProbe	159. 1693	2.01	0.45	47.22	0.02	0.02	0.02	157.29	58.91	27.38	5.42	-6.57	-44.07
LOOP6	P2	GeoProbe	159. 1693	2.005	0.45	47.22	0.02	0.02	0.02	157.29	58.91	27.38	5.42	-6.57	-44.07
LOOP6	P2	GeoProbe	159. 1693	2.40	1.25	31.39	0.09	0.06	0.02	83.07	383.83	41.12	22.71	-7.12	-46.85
LOOP6	P2	GeoProbe	159. 1693	2.395	1.25	31.39	0.09	0.06	0.02	83.07	383.83	41.12	22.71	-7.12	-46.85
LOOP6	P2	GeoProbe	159. 1693	3.62	0.51	23.26	0.02	0.16	0.02	76.71					
LOOP6	P2	GeoProbe	159. 1693	3.615	0.51	23.26	0.02	0.16	0.02	76.71					
LOOP6	P2	GeoProbe	159. 1693	4.84	0.56	21.90	0.07	0.92	0.02	56.84	187.37	38.27	36.81	-7.07	-47.35
LOOP6	P2	GeoProbe	159. 1693	4.835	0.56	21.90	0.07	0.92	0.02	56.84	187.37	38.27	36.81	-7.07	-47.35
LOOP6	P2A	GeoProbe	159. 1694	2.05	0.34	14.56	0.02	0.02	0.02	66.26	87.80	42.43	51.70	-6.36	-42.24
LOOP6	P2A	GeoProbe	159. 1694	2.045	0.34	14.56	0.02	0.02	0.02	66.26	87.80	42.43	51.70	-6.36	-42.24
LOOP6	P2A	GeoProbe	159. 1694	2.40	0.56	30.82	0.11	0.02	0.02	216.34	173.74	44.65	41.89	-6.74	-44.09
LOOP6	P2A	GeoProbe	159. 1694	2.395	0.56	30.82	0.11	0.02	0.02	216.34	173.74	44.65	41.89	-6.74	-44.09
LOOP6	P2A	GeoProbe	159. 1694	2.49	0.88	30.01	0.08	0.02	0.02	51.37	396.27	33.25	51.99	-7.06	-46.30
LOOP6	P2A	GeoProbe	159. 1694	2.485	0.88	30.01	0.08	0.02	0.02	51.37	396.27	33.25	51.99	-7.06	-46.30
LOOP6	P2A	GeoProbe	159. 1694	3.62	1.10	34.15	0.14	0.17	0.02	67.22	623.77	48.56	42.82	-6.85	-45.57
LOOP6	P2A	GeoProbe	159. 1694	3.615	1.10	34.15	0.14	0.17	0.02	67.22	623.77	48.56	42.82	-6.85	-45.57
LOOP6	P2A	GeoProbe	159. 1694	4.84	0.65	23.60	0.02	0.13	0.02	62.36	395.39	48.66	37.55	-6.62	-45.23
LOOP6	P2A	GeoProbe	159. 1694	4.835	0.65	23.60	0.02	0.13	0.02	62.36	395.39	48.66	37.55	-6.62	-45.23
LOOP6	P3	GeoProbe	159. 1695	2.40	0.19	56.85	0.18	128.58	0.30	20.62	298.44	32.48	2.11	-7.02	-48.58
LOOP6	P3	GeoProbe	159. 1695	2.395	0.19	56.85	0.18	128.58	0.30	20.62	298.44	32.48	2.11	-7.02	-48.58

LOOP6	P3	GeoProbe	159. 1695	3.62	0.45	20.76	0.09	89.45	0.33	16.50	378.32	51.85	0.86	-7.38	-50.08
LOOP6	P3	GeoProbe	159. 1695	3.615	0.45	20.76	0.09	89.45	0.33	16.50	378.32	51.85	0.86	-7.38	-50.08
LOOP6	P3	GeoProbe	159. 1695	4.84	0.10	17.67	0.11	55.92	0.02	28.69	129.29	21.93	2.05	-6.92	-44.47
LOOP6	P3	GeoProbe	159. 1695	4.835	0.10	17.67	0.11	55.92	0.02	28.69	129.29	21.93	2.05	-6.92	-44.47
LOOP6	P3	GeoProbe	159. 1695	6.01	0.34	20.77	0.09	42.88	0.02	61.93	19.42		48.81		
LOOP6	P3	GeoProbe	159. 1695	6.010	0.34	20.77	0.09	42.88	0.02	61.93	19.42		48.81		
LOOP6	P3	GeoProbe	159. 1695	6.51	0.49	17.44	0.09	58.29	0.02	24.39	39.83	32.46	4.76	-6.78	-44.55
LOOP6	P3	GeoProbe	159. 1695	6.505	0.49	17.44	0.09	58.29	0.02	24.39	39.83	32.46	4.76	-6.78	-44.55
LOOP6	P3	GeoProbe	159. 1695	6.73	0.51	23.71	0.09	45.28	0.02	42.66	44.66	36.07	27.85	-7.14	-47.41
LOOP6	P3	GeoProbe	159. 1695	6.725	0.51	23.71	0.09	45.28	0.02	42.66	44.66	36.07	27.85	-7.14	-47.41
LOOP6	P3	GeoProbe	159. 1695	7.18	0.29	24.61	0.13	0.68	0.02	64.84	62.91	32.56	20.27	-7.72	-51.26
LOOP6	P3	GeoProbe	159. 1695	7.175	0.29	24.61	0.13	0.68	0.02	64.84	62.91	32.56	20.27	-7.72	-51.26
LOOP6	P3	GeoProbe	159. 1695	8.06	1.38	26.16	0.12	0.14	0.02	79.02	72.50		10.69		
LOOP6	P3	GeoProbe	159. 1695	8.055	1.38	26.16	0.12	0.14	0.02	79.02	72.50		10.69		
LOOP6	P3	GeoProbe	159. 1695	8.50	1.03	24.38	0.11	0.39	0.17	62.42	116.90	59.94	7.44		
LOOP6	P3	GeoProbe	159. 1695	8.495	1.03	24.38	0.11	0.39	0.17	62.42	116.90	59.94	7.44		
LOOP6	P3	GeoProbe	159. 1695	9.03	0.51	20.85	0.15	0.06	0.02	39.61	117.06	35.41	14.70	-7.87	-52.38
LOOP6	P3	GeoProbe	159. 1695	9.025	0.51	20.85	0.15	0.06	0.02	39.61	117.06	35.41	14.70	-7.87	-52.38
LOOP6	P3	GeoProbe	159. 1695	9.72	0.98	27.19	0.12	0.29	0.38	58.27	225.03	53.39	15.07		
LOOP6	P3	GeoProbe	159. 1695	9.715	0.98	27.19	0.12	0.29	0.38	58.27	225.03	53.39	15.07		
LOOP6	P4	GeoProbe	159. 1696	1.71	0.67	34.27	0.11	121.23	0.07	19.77	156.71	37.01	1.61	-6.82	-44.69
LOOP6	P4	GeoProbe	159. 1696	1.705	0.67	34.27	0.11	121.23	0.07	19.77	156.71	37.01	1.61	-6.82	-44.69
LOOP6	P4	GeoProbe	159. 1696	2.40	0.15	57.14	0.16	240.06	0.02	11.61	72.62	69.36	3.69	-5.95	-40.61
LOOP6	P4	GeoProbe	159. 1696	2.395	0.15	57.14	0.16	240.06	0.02	11.61	72.62	69.36	3.69	-5.95	-40.61
LOOP6	P4	GeoProbe	159. 1696	4.84	0.72	33.30	0.12	70.67	0.02	18.50	110.86	36.05	6.89	-7.14	-45.57
LOOP6	P4	GeoProbe	159. 1696	4.835	0.72	33.30	0.12	70.67	0.02	18.50	110.86	36.05	6.89	-7.14	-45.57
LOOP6	P4	GeoProbe	159. 1696	6.06	0.72	28.73	0.11	83.36	0.02	28.63	104.33	69.63	7.56		
LOOP6	P4	GeoProbe	159. 1696	6.055	0.72	28.73	0.11	83.36	0.02	28.63	104.33	69.63	7.56		
LOOP6	P4	GeoProbe	159. 1696	6.19	0.47	21.40	0.13	0.09	0.02	67.18	143.81	50.97	5.76	-7.11	-45.70
LOOP6	P4	GeoProbe	159. 1696	6.185	0.47	21.40	0.13	0.09	0.02	67.18	143.81	50.97	5.76	-7.11	-45.70
LOOP6	P4	GeoProbe	159. 1696	7.28	1.38	57.14	0.40	0.26	0.15	106.18	289.75	72.23	25.88		
LOOP6	P4	GeoProbe	159. 1696	7.275	1.38	57.14	0.40	0.26	0.15	106.18	289.75	72.23	25.88		
LOOP6	P5	GeoProbe	159. 1697	2.40	0.48	34.88	0.21	69.19	0.02	20.24	80.68	37.49	4.01	-7.13	-46.00
LOOP6	P5	GeoProbe	159. 1697	2.395	0.48	34.88	0.21	69.19	0.02	20.24	80.68	37.49	4.01	-7.13	-46.00

LOOP6	P5	GeoProbe	159. 1697	3.62	0.17	37.14	0.11	71.49	0.02	38.73	72.98	29.05	6.58	-6.84	-43.35
LOOP6	P5	GeoProbe	159. 1697	3.615	0.17	37.14	0.11	71.49	0.02	38.73	72.98	29.05	6.58	-6.84	-43.35
LOOP6	P5	GeoProbe	159. 1697	4.84	0.60	15.06	0.02	47.53	0.02	29.90	39.24	105.62	4.47		
LOOP6	P5	GeoProbe	159. 1697	4.835	0.60	15.06	0.02	47.53	0.02	29.90	39.24	105.62	4.47		
LOOP6	P5	GeoProbe	159. 1697	6.06	0.62	21.53	0.02	120.41	0.02	12.27	265.37	92.98	3.10	-7.17	-46.77
LOOP6	P5	GeoProbe	159. 1697	6.055	0.62	21.53	0.02	120.41	0.02	12.27	265.37	92.98	3.10	-7.17	-46.77
LOOP6	P5	GeoProbe	159. 1697	7.16	0.05	40.03	0.09	52.32	0.02	26.74	53.32	31.17	6.32	-7.05	-46.68
LOOP6	P5	GeoProbe	159. 1697	7.155	0.05	40.03	0.09	52.32	0.02	26.74	53.32	31.17	6.32	-7.05	-46.68
LOOP6	P5	GeoProbe	159. 1697	7.72	0.07	45.39	0.02	7.93	0.02	43.76	11.34	33.07	9.29	-7.21	-47.64
LOOP6	P5	GeoProbe	159. 1697	7.715	0.07	45.39	0.02	7.93	0.02	43.76	11.34	33.07	9.29	-7.21	-47.64
LOOP6	P5	GeoProbe	159. 1697	8.44	0.08	81.58	0.19	0.02	0.02	50.81	336.85	42.46	8.27	-7.27	-47.88
LOOP6	P5	GeoProbe	159. 1697	8.435	0.08	81.58	0.19	0.02	0.02	50.81	336.85	42.46	8.27	-7.27	-47.88
LOOP6	P5	GeoProbe	159. 1697	9.61	0.31	33.19	0.06	0.02	0.02	61.95	803.49	41.30	19.75		
LOOP6	P5	GeoProbe	159. 1697	9.610	0.31	33.19	0.06	0.02	0.02	61.95	803.49	41.30	19.75		
LOOP6	P5	GeoProbe	159. 1697	10.94	0.20	32.16	0.16	0.05	0.02	63.67	809.51	47.62	22.43	-7.54	-50.15
LOOP6	P5	GeoProbe	159. 1697	10.935	0.20	32.16	0.16	0.05	0.02	63.67	809.51	47.62	22.43	-7.54	-50.15
LOOP6	P6	GeoProbe	159. 1698	2.40	0.10	21.55	0.10	39.08	0.02	31.29	3.15	28.11	19.32	-7.26	-46.46
LOOP6	P6	GeoProbe	159. 1698	2.395	0.10	21.55	0.10	39.08	0.02	31.29	3.15	28.11	19.32	-7.26	-46.46
LOOP6	P6	GeoProbe	159. 1698	3.34	0.28	23.67	0.15	35.39	0.16	31.73	32.13	25.07	16.42	-7.19	-46.00
LOOP6	P6	GeoProbe	159. 1698	3.340	0.28	23.67	0.15	35.39	0.16	31.73	32.13	25.07	16.42	-7.19	-46.00
LOOP6	P6	GeoProbe	159. 1698	3.57	0.21	24.53	0.15	28.55	0.02	31.86	38.68	29.60		-7.28	-46.65
LOOP6	P6	GeoProbe	159. 1698	3.570	0.21	24.53	0.15	28.55	0.02	31.86	38.68	29.60		-7.28	-46.65
LOOP6	P6	GeoProbe	159. 1698	4.74	0.15	18.58	0.08	25.98	0.02	28.63	45.39	14.74	14.03	-7.18	-45.47
LOOP6	P6	GeoProbe	159. 1698	4.735	0.15	18.58	0.08	25.98	0.02	28.63	45.39	14.74	14.03	-7.18	-45.47
LOOP6	P6	GeoProbe	159. 1698	5.67	0.19	31.37	0.09	5.10	0.02	49.72	12.45	42.52		-7.18	-47.24
LOOP6	P6	GeoProbe	159. 1698	5.670	0.19	31.37	0.09	5.10	0.02	49.72	12.45	42.52		-7.18	-47.24
LOOP6	P6	GeoProbe	159. 1698	6.02	0.25	37.43	0.07	2.10	0.02	61.89	16.37	25.85	41.86	-7.52	-48.62
LOOP6	P6	GeoProbe	159. 1698	6.015	0.25	37.43	0.07	2.10	0.02	61.89	16.37	25.85	41.86	-7.52	-48.62
LOOP6	P6	GeoProbe	159. 1698	6.64	2.25	43.31	0.14	0.32	0.05	65.15	178.14				
LOOP6	P6	GeoProbe	159. 1698	6.635	2.25	43.31	0.14	0.32	0.05	65.15	178.14				
LOOP6	P6	GeoProbe	159. 1698	8.50	1.51	19.96	0.07	0.46	0.12	52.19	566.46	46.44	20.35	-7.57	-49.94
LOOP6	P6	GeoProbe	159. 1698	8.495	1.51	19.96	0.07	0.46	0.12	52.19	566.46	46.44	20.35	-7.57	-49.94
LOOP6	P7	GeoProbe	159. 1699	2.31	0.45	45.70	0.10	158.19	0.02	17.77	879.00	40.88	2.22		
LOOP6	P7	GeoProbe	159. 1699	2.305	0.45	45.70	0.10	158.19	0.02	17.77	879.00	40.88	2.22		

LOOP6	P7	GeoProbe	159. 1699	3.31		25.40	0.09	95.74	0.02	72.53	146.58	51.89	35.47	-7.17	-45.60
LOOP6	P7	GeoProbe	159. 1699	3.305		25.40	0.09	95.74	0.02	72.53	146.58	51.89	35.47	-7.17	-45.60
LOOP6	P7	GeoProbe	159. 1699	4.84	0.13	24.79	0.17	94.82	0.02	24.93	300.66	27.31	4.73	-7.28	-46.08
LOOP6	P7	GeoProbe	159. 1699	4.835	0.13	24.79	0.17	94.82	0.02	24.93	300.66	27.31	4.73	-7.28	-46.08
LOOP6	P7	GeoProbe	159. 1699	6.06	0.33	23.18	0.14	98.97	0.02	20.38	36.44	58.10	34.53	-7.27	-47.06
LOOP6	P7	GeoProbe	159. 1699	6.055	0.33	23.18	0.14	98.97	0.02	20.38	36.44	58.10	34.53	-7.27	-47.06
LOOP6	P7	GeoProbe	159. 1699	7.13	0.48	27.36	0.18	15.26	0.02	32.74	813.55	48.61	33.83	-7.19	-46.62
LOOP6	P7	GeoProbe	159. 1699	7.125	0.48	27.36	0.18	15.26	0.02	32.74	813.55	48.61	33.83	-7.19	-46.62
LOOP6	P7	GeoProbe	159. 1699	8.47	0.23	21.11	0.08	58.17	0.02	17.51	41.56	47.95	37.60	-7.19	-46.91
LOOP6	P7	GeoProbe	159. 1699	8.465	0.23	21.11	0.08	58.17	0.02	17.51	41.56	47.95	37.60	-7.19	-46.91
LOOP6	P8	GeoProbe	159. 1700	1.98	0.08	15.88	0.06	5.91	0.02	33.40	36.96	28.02	4.02	-6.79	-44.48
LOOP6	P8	GeoProbe	159. 1700	1.975	0.08	15.88	0.06	5.91	0.02	33.40	36.96	28.02	4.02	-6.79	-44.48
LOOP6	P8	GeoProbe	159. 1700	2.40	0.24	9.79	0.02	0.02	0.02	146.87	189.05	31.85	2.58	-6.67	-43.53
LOOP6	P8	GeoProbe	159. 1700	2.395	0.24	9.79	0.02	0.02	0.02	146.87	189.05	31.85	2.58	-6.67	-43.53
LOOP6	P8	GeoProbe	159. 1700	3.52	0.06	12.40	0.10	0.09	0.02	74.49		53.87	35.39		
LOOP6	P8	GeoProbe	159. 1700	3.515	0.06	12.40	0.10	0.09	0.02	74.49		53.87	35.39		
LOOP6	P8	GeoProbe	159. 1700	3.62	0.29	20.44	0.12	48.63	0.02	13.30	130.77	23.98	5.06	-6.84	-44.58
LOOP6	P8	GeoProbe	159. 1700	3.615	0.29	20.44	0.12	48.63	0.02	13.30	130.77	23.98	5.06	-6.84	-44.58
LOOP6	P8	GeoProbe	159. 1700	4.74	0.31	26.61	0.09	0.02	0.02	86.18	816.50	56.32	35.82	-6.62	-45.49
LOOP6	P8	GeoProbe	159. 1700	4.735	0.31	26.61	0.09	0.02	0.02	86.18	816.50	56.32	35.82	-6.62	-45.49
LOOP6	P8	GeoProbe	159. 1700	5.96	0.42	22.91	0.08	0.02	0.02	93.37	740.83	47.01	31.58		
LOOP6	P8	GeoProbe	159. 1700	5.955	0.42	22.91	0.08	0.02	0.02	93.37	740.83	47.01	31.58		
LOOP6	P8	GeoProbe	159. 1700	7.18	0.16	21.87	0.07	0.02	0.02	88.34	56.18	46.86			
LOOP6	P8	GeoProbe	159. 1700	7.175	0.16	21.87	0.07	0.02	0.02	88.34	56.18	46.86			

Appendix3-2. Results of cations of LOOP6

LOOPNr	ID	method	DGUnr	Depth (m)	mg/									
					Al	Ba	Ca	Fe	K	Mg	Mn	Na	Ni	Sr
LOOP6	P1	GeoProbe	159. 1692	1.95	0.15	0.09	58.77	0.03	13.63	15.16	0.09	39.69	0.01	0.27
LOOP6	P1	GeoProbe	159. 1692	1.945	0.15	0.09	58.77	0.03	13.63	15.16	0.09	39.69	0.01	0.27
LOOP6	P1	GeoProbe	159. 1692	2.41	0.15	0.13	63.27	0.19	7.17	9.25	0.28	32.30	0.09	0.35

LOOP6	P1	GeoProbe	159. 1692	2.405	0.15	0.13	63.27	0.19	7.17	9.25	0.28	32.30	0.09	0.35
LOOP6	P1	GeoProbe	159. 1692	3.62	0.15	0.12	49.15	0.11	12.35	6.69	0.46	24.56	0.03	0.29
LOOP6	P1	GeoProbe	159. 1692	3.615	0.15	0.12	49.15	0.11	12.35	6.69	0.46	24.56	0.03	0.29
LOOP6	P1	GeoProbe	159. 1692	4.84	0.13	0.11	88.25	0.03	6.78	7.81	0.34	34.45	0.03	0.31
LOOP6	P1	GeoProbe	159. 1692	4.835	0.13	0.11	88.25	0.03	6.78	7.81	0.34	34.45	0.03	0.31
LOOP6	P1	GeoProbe	159. 1692	6.06	0.13	0.10	74.39	0.03	2.20	5.74	0.36	19.60	0.01	0.24
LOOP6	P1	GeoProbe	159. 1692	6.055	0.13	0.10	74.39	0.03	2.20	5.74	0.36	19.60	0.01	0.24
LOOP6	P2	GeoProbe	159. 1693	2.01	0.20	0.08	66.79	0.09	2.38	3.77	0.13	36.09	0.13	0.22
LOOP6	P2	GeoProbe	159. 1693	2.005	0.20	0.08	66.79	0.09	2.38	3.77	0.13	36.09	0.13	0.22
LOOP6	P2	GeoProbe	159. 1693	2.40	0.16	0.15	66.08	0.10	10.09	6.78	0.79	20.69	0.02	0.27
LOOP6	P2	GeoProbe	159. 1693	2.395	0.16	0.15	66.08	0.10	10.09	6.78	0.79	20.69	0.02	0.27
LOOP6	P2	GeoProbe	159. 1693	4.84	0.15	0.09	73.43	0.03	2.87	5.67	0.37	17.49	0.01	0.23
LOOP6	P2	GeoProbe	159. 1693	4.835	0.15	0.09	73.43	0.03	2.87	5.67	0.37	17.49	0.01	0.23
LOOP6	P2A	GeoProbe	159. 1694	2.05	0.21	0.07	107.31	0.30	1.99	4.42	0.13	13.20	0.09	0.26
LOOP6	P2A	GeoProbe	159. 1694	2.045	0.21	0.07	107.31	0.30	1.99	4.42	0.13	13.20	0.09	0.26
LOOP6	P2A	GeoProbe	159. 1694	2.40	0.14	0.17	155.19	0.10	5.07	6.31	0.90	17.80	0.05	0.38
LOOP6	P2A	GeoProbe	159. 1694	2.395	0.14	0.17	155.19	0.10	5.07	6.31	0.90	17.80	0.05	0.38
LOOP6	P2A	GeoProbe	159. 1694	2.49	0.20	0.14	105.40	0.90	3.53	5.49	1.54	13.87	0.03	0.30
LOOP6	P2A	GeoProbe	159. 1694	2.485	0.20	0.14	105.40	0.90	3.53	5.49	1.54	13.87	0.03	0.30
LOOP6	P2A	GeoProbe	159. 1694	3.62	0.14	0.08	92.53	0.45	2.61	5.14	0.78	18.33	0.01	0.21
LOOP6	P2A	GeoProbe	159. 1694	3.615	0.14	0.08	92.53	0.45	2.61	5.14	0.78	18.33	0.01	0.21
LOOP6	P2A	GeoProbe	159. 1694	4.84	0.20	0.09	90.56	0.63	2.09	5.50	0.81	16.99	0.09	0.21
LOOP6	P2A	GeoProbe	159. 1694	4.835	0.20	0.09	90.56	0.63	2.09	5.50	0.81	16.99	0.09	0.21
LOOP6	P3	GeoProbe	159. 1695	2.40	0.17	0.05	57.87	0.12	13.17	8.20	0.47	22.97	0.58	0.14
LOOP6	P3	GeoProbe	159. 1695	2.395	0.17	0.05	57.87	0.12	13.17	8.20	0.47	22.97	0.58	0.14
LOOP6	P3	GeoProbe	159. 1695	3.62	0.19	0.02	25.71	0.16	14.06	5.44	0.22	20.26	0.02	0.08
LOOP6	P3	GeoProbe	159. 1695	3.615	0.19	0.02	25.71	0.16	14.06	5.44	0.22	20.26	0.02	0.08
LOOP6	P3	GeoProbe	159. 1695	4.84	0.14	0.07	27.01	0.10	9.57	3.66	0.45	14.38	0.03	0.14
LOOP6	P3	GeoProbe	159. 1695	4.835	0.14	0.07	27.01	0.10	9.57	3.66	0.45	14.38	0.03	0.14
LOOP6	P3	GeoProbe	159. 1695	6.01	0.17	0.04	98.08	0.16	3.78	10.17	0.02	23.39	0.01	0.35
LOOP6	P3	GeoProbe	159. 1695	6.010	0.17	0.04	98.08	0.16	3.78	10.17	0.02	23.39	0.01	0.35

LOOP6	P3	GeoProbe	159. 1695	6.51	0.15	0.11	30.28	0.10	3.71	5.29	0.05	19.72	0.01	0.20
LOOP6	P3	GeoProbe	159. 1695	6.505	0.15	0.11	30.28	0.10	3.71	5.29	0.05	19.72	0.01	0.20
LOOP6	P3	GeoProbe	159. 1695	6.73	0.16	0.08	67.21	0.15	2.34	7.73	0.02	25.75	0.01	0.33
LOOP6	P3	GeoProbe	159. 1695	6.725	0.16	0.08	67.21	0.15	2.34	7.73	0.02	25.75	0.01	0.33
LOOP6	P3	GeoProbe	159. 1695	7.18	0.15	0.03	56.97	0.17	1.89	4.61	0.03	20.04	0.01	0.18
LOOP6	P3	GeoProbe	159. 1695	7.175	0.15	0.03	56.97	0.17	1.89	4.61	0.03	20.04	0.01	0.18
LOOP6	P3	GeoProbe	159. 1695	8.06	0.17	0.08	44.42	0.03	1.37	4.09	0.15	31.41	0.00	0.16
LOOP6	P3	GeoProbe	159. 1695	8.055	0.17	0.08	44.42	0.03	1.37	4.09	0.15	31.41	0.00	0.16
LOOP6	P3	GeoProbe	159. 1695	8.50	0.17	0.05	35.72	0.03	1.82	3.29	0.07	20.94	0.00	0.14
LOOP6	P3	GeoProbe	159. 1695	8.495	0.17	0.05	35.72	0.03	1.82	3.29	0.07	20.94	0.00	0.14
LOOP6	P3	GeoProbe	159. 1695	9.03	0.16	0.05	40.51	0.03	1.91	3.89	0.10	14.43	0.01	0.14
LOOP6	P3	GeoProbe	159. 1695	9.025	0.16	0.05	40.51	0.03	1.91	3.89	0.10	14.43	0.01	0.14
LOOP6	P3	GeoProbe	159. 1695	9.72	0.15	0.07	46.10	0.03	2.62	5.76	0.07	26.39	0.01	0.29
LOOP6	P3	GeoProbe	159. 1695	9.715	0.15	0.07	46.10	0.03	2.62	5.76	0.07	26.39	0.01	0.29
LOOP6	P4	GeoProbe	159. 1696	1.71	0.35	0.19	38.94	0.03	4.61	5.30	0.12	35.79	0.01	0.27
LOOP6	P4	GeoProbe	159. 1696	1.705	0.35	0.19	38.94	0.03	4.61	5.30	0.12	35.79	0.01	0.27
LOOP6	P4	GeoProbe	159. 1696	2.40	0.14	0.14	64.89	0.03	16.11	12.54	0.13	37.48	0.01	0.28
LOOP6	P4	GeoProbe	159. 1696	2.395	0.14	0.14	64.89	0.03	16.11	12.54	0.13	37.48	0.01	0.28
LOOP6	P4	GeoProbe	159. 1696	4.84	0.20	0.18	23.61	0.03	2.99	13.41	0.28	32.99	0.03	0.28
LOOP6	P4	GeoProbe	159. 1696	4.835	0.20	0.18	23.61	0.03	2.99	13.41	0.28	32.99	0.03	0.28
LOOP6	P4	GeoProbe	159. 1696	6.06	0.20	0.13	24.76	0.03	2.07	11.35	0.08	41.81	0.03	0.35
LOOP6	P4	GeoProbe	159. 1696	6.055	0.20	0.13	24.76	0.03	2.07	11.35	0.08	41.81	0.03	0.35
LOOP6	P4	GeoProbe	159. 1696	6.19	0.16	0.05	17.19	0.03	2.44	9.72	0.14	25.30	0.17	0.20
LOOP6	P4	GeoProbe	159. 1696	6.185	0.16	0.05	17.19	0.03	2.44	9.72	0.14	25.30	0.17	0.20
LOOP6	P4	GeoProbe	159. 1696	7.28	0.15	0.07	74.54	0.03	4.83	13.09	0.22	41.80	0.04	0.47
LOOP6	P4	GeoProbe	159. 1696	7.275	0.15	0.07	74.54	0.03	4.83	13.09	0.22	41.80	0.04	0.47
LOOP6	P5	GeoProbe	159. 1697	2.40	0.34	0.15	19.67	0.03	3.21	7.29	0.62	36.51	0.03	0.15
LOOP6	P5	GeoProbe	159. 1697	2.395	0.34	0.15	19.67	0.03	3.21	7.29	0.62	36.51	0.03	0.15
LOOP6	P5	GeoProbe	159. 1697	3.62	0.24	0.04	41.50	0.03	12.28	5.77	0.83	22.60	0.02	0.11
LOOP6	P5	GeoProbe	159. 1697	3.615	0.24	0.04	41.50	0.03	12.28	5.77	0.83	22.60	0.02	0.11
LOOP6	P5	GeoProbe	159. 1697	4.84	0.16	0.07	22.94	0.03	7.46	4.85	0.03	27.42	0.00	0.09

LOOP6	P5	GeoProbe	159. 1697	4.835	0.16	0.07	22.94	0.03	7.46	4.85	0.03	27.42	0.00	0.09
LOOP6	P5	GeoProbe	159. 1697	6.06	0.17	0.12	33.55	0.03	10.54	3.22	1.22	22.95	0.06	0.13
LOOP6	P5	GeoProbe	159. 1697	6.055	0.17	0.12	33.55	0.03	10.54	3.22	1.22	22.95	0.06	0.13
LOOP6	P5	GeoProbe	159. 1697	7.16	0.13	0.07	38.98	0.03	6.24	4.39	0.15	26.08	0.01	0.14
LOOP6	P5	GeoProbe	159. 1697	7.155	0.13	0.07	38.98	0.03	6.24	4.39	0.15	26.08	0.01	0.14
LOOP6	P5	GeoProbe	159. 1697	7.72	0.13	0.06	32.69	0.03	3.41	6.12	0.53	33.16	0.05	0.18
LOOP6	P5	GeoProbe	159. 1697	7.715	0.13	0.06	32.69	0.03	3.41	6.12	0.53	33.16	0.05	0.18
LOOP6	P5	GeoProbe	159. 1697	8.44	0.14	0.07	31.17	0.03	3.09	4.84	0.11	53.69	0.05	0.15
LOOP6	P5	GeoProbe	159. 1697	8.435	0.14	0.07	31.17	0.03	3.09	4.84	0.11	53.69	0.05	0.15
LOOP6	P5	GeoProbe	159. 1697	9.61	0.11	0.12	45.53	1.00	4.47	9.41	0.85	19.11	0.01	0.21
LOOP6	P5	GeoProbe	159. 1697	9.610	0.11	0.12	45.53	1.00	4.47	9.41	0.85	19.11	0.01	0.21
LOOP6	P5	GeoProbe	159. 1697	10.94	0.11	0.10	43.51	0.58	6.81	8.86	0.77	24.77	0.01	0.21
LOOP6	P5	GeoProbe	159. 1697	10.935	0.11	0.10	43.51	0.58	6.81	8.86	0.77	24.77	0.01	0.21
LOOP6	P6	GeoProbe	159. 1698	2.40	0.18	0.10	42.46	0.03	11.54	8.39	0.03	13.67	0.01	0.13
LOOP6	P6	GeoProbe	159. 1698	2.395	0.18	0.10	42.46	0.03	11.54	8.39	0.03	13.67	0.01	0.13
LOOP6	P6	GeoProbe	159. 1698	3.34	0.23	0.15	44.05	0.03	8.18	8.01	0.05	18.87	0.01	0.18
LOOP6	P6	GeoProbe	159. 1698	3.340	0.23	0.15	44.05	0.03	8.18	8.01	0.05	18.87	0.01	0.18
LOOP6	P6	GeoProbe	159. 1698	3.57	0.14	0.08	40.26	0.03	5.09	7.67	0.28	17.99	0.01	0.17
LOOP6	P6	GeoProbe	159. 1698	3.570	0.14	0.08	40.26	0.03	5.09	7.67	0.28	17.99	0.01	0.17
LOOP6	P6	GeoProbe	159. 1698	4.74	0.13	0.05	36.72	0.03	2.40	7.64	0.02	15.19	0.00	0.17
LOOP6	P6	GeoProbe	159. 1698	4.735	0.13	0.05	36.72	0.03	2.40	7.64	0.02	15.19	0.00	0.17
LOOP6	P6	GeoProbe	159. 1698	5.67	0.17	0.06	87.91	0.03	2.82	8.10	0.08	23.24	0.01	0.25
LOOP6	P6	GeoProbe	159. 1698	5.670	0.17	0.06	87.91	0.03	2.82	8.10	0.08	23.24	0.01	0.25
LOOP6	P6	GeoProbe	159. 1698	6.02	0.15	0.08	82.44	0.03	3.42	7.17	0.16	28.42	0.01	0.27
LOOP6	P6	GeoProbe	159. 1698	6.015	0.15	0.08	82.44	0.03	3.42	7.17	0.16	28.42	0.01	0.27
LOOP6	P6	GeoProbe	159. 1698	6.64	0.14	0.08	62.28	0.03	3.50	7.56	0.08	44.51	0.02	0.52
LOOP6	P6	GeoProbe	159. 1698	6.635	0.14	0.08	62.28	0.03	3.50	7.56	0.08	44.51	0.02	0.52
LOOP6	P6	GeoProbe	159. 1698	7.18	0.72	0.22	58.05	0.69	7.59	7.35	0.11	60.65	0.04	0.43
LOOP6	P6	GeoProbe	159. 1698	7.175	0.72	0.22	58.05	0.69	7.59	7.35	0.11	60.65	0.04	0.43
LOOP6	P6	GeoProbe	159. 1698	8.50	0.13	0.12	45.00	0.54	7.47	7.70	0.07	24.56	0.02	0.48
LOOP6	P6	GeoProbe	159. 1698	8.495	0.13	0.12	45.00	0.54	7.47	7.70	0.07	24.56	0.02	0.48

LOOP6	P7	GeoProbe	159. 1699	2.31	0.44	0.16	41.13	0.13	17.52	6.96	10.60	28.11	0.38	0.14
LOOP6	P7	GeoProbe	159. 1699	2.305	0.44	0.16	41.13	0.13	17.52	6.96	10.60	28.11	0.38	0.14
LOOP6	P7	GeoProbe	159. 1699	3.31	0.16	0.06	113.73	0.03	3.66	7.17	0.94	10.64	0.04	0.18
LOOP6	P7	GeoProbe	159. 1699	3.305	0.16	0.06	113.73	0.03	3.66	7.17	0.94	10.64	0.04	0.18
LOOP6	P7	GeoProbe	159. 1699	4.84	0.15	0.08	42.51	0.03	4.39	3.16	1.91	19.20	0.07	0.20
LOOP6	P7	GeoProbe	159. 1699	4.835	0.15	0.08	42.51	0.03	4.39	3.16	1.91	19.20	0.07	0.20
LOOP6	P7	GeoProbe	159. 1699	6.06	0.10	0.03	88.65	0.03	2.44	6.51	0.03	18.69	0.01	0.43
LOOP6	P7	GeoProbe	159. 1699	6.055	0.10	0.03	88.65	0.03	2.44	6.51	0.03	18.69	0.01	0.43
LOOP6	P7	GeoProbe	159. 1699	7.13	0.15	0.05	58.74	0.03	3.65	5.34	1.12	25.71	0.01	0.30
LOOP6	P7	GeoProbe	159. 1699	7.125	0.15	0.05	58.74	0.03	3.65	5.34	1.12	25.71	0.01	0.30
LOOP6	P7	GeoProbe	159. 1699	8.47	0.11	0.03	77.56	0.03	2.50	6.58	0.02	18.24	0.01	0.43
LOOP6	P7	GeoProbe	159. 1699	8.465	0.11	0.03	77.56	0.03	2.50	6.58	0.02	18.24	0.01	0.43
LOOP6	P8	GeoProbe	159. 1700	1.98	0.53	0.06	17.77	0.03	5.57	1.88	0.06	15.83	0.03	0.06
LOOP6	P8	GeoProbe	159. 1700	1.975	0.53	0.06	17.77	0.03	5.57	1.88	0.06	15.83	0.03	0.06
LOOP6	P8	GeoProbe	159. 1700	2.40	0.46	0.06	43.20	3.85	10.43	4.03	0.17	17.63	0.43	0.14
LOOP6	P8	GeoProbe	159. 1700	2.395	0.46	0.06	43.20	3.85	10.43	4.03	0.17	17.63	0.43	0.14
LOOP6	P8	GeoProbe	159. 1700	3.52	0.41	0.07	24.72	1.13	10.49	2.47	0.18	19.98	0.18	0.09
LOOP6	P8	GeoProbe	159. 1700	3.515	0.41	0.07	24.72	1.13	10.49	2.47	0.18	19.98	0.18	0.09
LOOP6	P8	GeoProbe	159. 1700	3.62	0.20	0.20	20.96	0.03	2.60	9.14	0.19	31.96	0.01	0.25
LOOP6	P8	GeoProbe	159. 1700	3.615	0.20	0.20	20.96	0.03	2.60	9.14	0.19	31.96	0.01	0.25
LOOP6	P8	GeoProbe	159. 1700	4.74	0.30	0.09	81.83	4.73	2.69	7.29	7.24	24.84	0.01	0.21
LOOP6	P8	GeoProbe	159. 1700	4.735	0.30	0.09	81.83	4.73	2.69	7.29	7.24	24.84	0.01	0.21
LOOP6	P8	GeoProbe	159. 1700	5.96	0.78	0.08	82.24	1.79	2.69	6.02	3.67	27.05	0.01	0.22
LOOP6	P8	GeoProbe	159. 1700	5.955	0.78	0.08	82.24	1.79	2.69	6.02	3.67	27.05	0.01	0.22
LOOP6	P8	GeoProbe	159. 1700	7.18	0.68	0.06	83.92	0.26	1.79	4.96	0.80	27.38	0.01	0.21
LOOP6	P8	GeoProbe	159. 1700	7.175	0.68	0.06	83.92	0.26	1.79	4.96	0.80	27.38	0.01	0.21

Appendix3-3. Formic acid extractable Fe(II), Fe(total), and Fe(II)/Fe(total) of LOOP6

LOOPNr	DGUnr	ID	Depth (m)	mg/Kg		Fe(II)/Fe(total)
				Fe(II)	Fe(total)	
LOOP6	159. 1692	P1	2.2	326.4	476.1	0.69
LOOP6	159. 1692	P1	3.51	312.0	476.5	0.65
LOOP6	159. 1692	P1	4.485	339.4	396.9	0.86
LOOP6	159. 1692	P1	5.96	493.9	518.2	0.95
LOOP6	159. 1693	P2	1.915	1127.9	1321.9	0.85
LOOP6	159. 1693	P2	3.375	1447.2	1242.2	1.16
LOOP6	159. 1693	P2	4.49	1197.3	1285.1	0.93
LOOP6	159. 1694	P2a	2.14	107.0	212.5	0.50
LOOP6	159. 1695	P3	2.395	0.3	39.3	0.01
LOOP6	159. 1695	P3	3.36	0.6	104.6	0.01
LOOP6	159. 1695	P3	4.565	1.8	175.1	0.01
LOOP6	159. 1695	P3	5.705	1.3	209.2	0.01
LOOP6	159. 1695	P3	6.82	3.1	96.3	0.03
LOOP6	159. 1695	P3	7.71	26.6	201.1	0.13
LOOP6	159. 1695	P3	8.15	446.8	506.2	0.88
LOOP6	159. 1695	P3	9.37	808.0	911.4	0.89
LOOP6	159. 1696	P4	2.3	175.0	766.2	0.23
LOOP6	159. 1696	P4	3.52	3.4	119.6	0.03
LOOP6	159. 1696	P4	4.74	2.4	228.3	0.01
LOOP6	159. 1696	P4	5.96	38.2	162.5	0.23
LOOP6	159. 1696	P4	6.09	106.6	134.5	0.79
LOOP6	159. 1696	P4	7.18	258.1	288.6	0.89
LOOP6	159. 1697	P5	2.3	0.5	34.6	0.01
LOOP6	159. 1697	P5	3.52	0.3	23.6	0.01
LOOP6	159. 1697	P5	4.785	0.1	9.2	0.01
LOOP6	159. 1697	P5	5.96	0.2	19.0	0.01
LOOP6	159. 1697	P5	6.615	12.9	157.9	0.08
LOOP6	159. 1697	P5	6.69	0.2	34.8	0.01
LOOP6	159. 1697	P5	6.75	2.5	234.4	0.01
LOOP6	159. 1697	P5	7.79	0.6	54.0	0.01
LOOP6	159. 1697	P5	8.51	2.3	27.0	0.08
LOOP6	159. 1697	P5	9.705	26.4	38.0	0.69
LOOP6	159. 1698	P6	2.3	235.7	326.0	0.72
LOOP6	159. 1698	P6	3.6375	0.8	115.6	0.01
LOOP6	159. 1698	P6	4.83	0.3	105.3	0.00
LOOP6	159. 1698	P6	6.08	163.3	302.9	0.54
LOOP6	159. 1698	P6	6.54	480.8	606.5	0.79
LOOP6	159. 1698	P6	7.27	516.7	648.9	0.80
LOOP6	159. 1699	P7	2.395	1.6	158.8	0.01

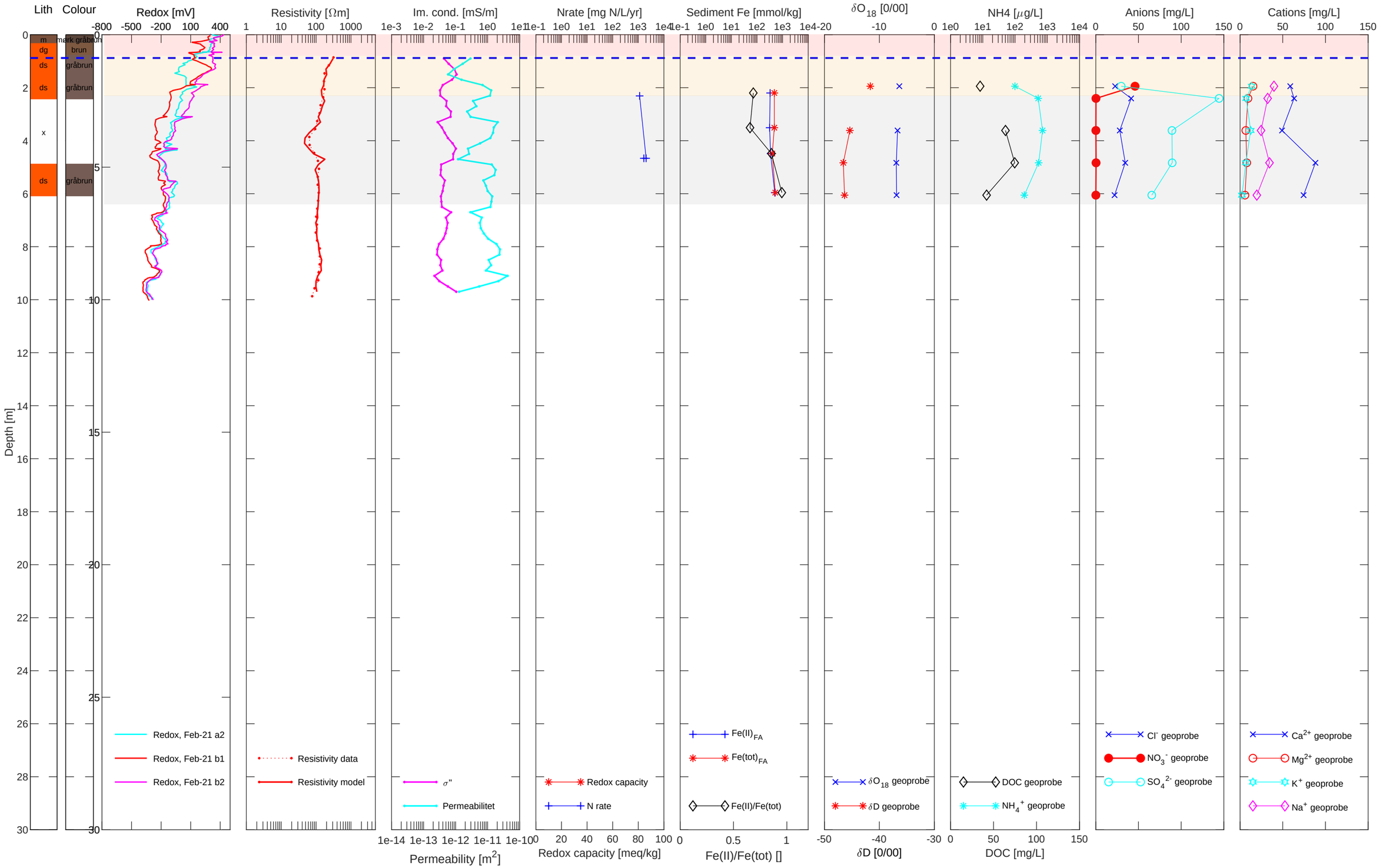
LOOP6	159. 1699	P7	5.96	1.4	213.1	0.01
LOOP6	159. 1699	P7	8.525	2.7	110.9	0.02
LOOP6	159. 1700	P8	1.88	3.5	61.7	0.06
LOOP6	159. 1700	P8	3.61	83.5	134.4	0.62
LOOP6	159. 1700	P8	4.83	531.8	595.1	0.89
LOOP6	159. 1700	P8	6.05	891.4	901.2	0.99
LOOP6	159. 1700	P8	7.27	114.3	254.1	0.45

Appendix3-4. Denitrification rate of LOOP6

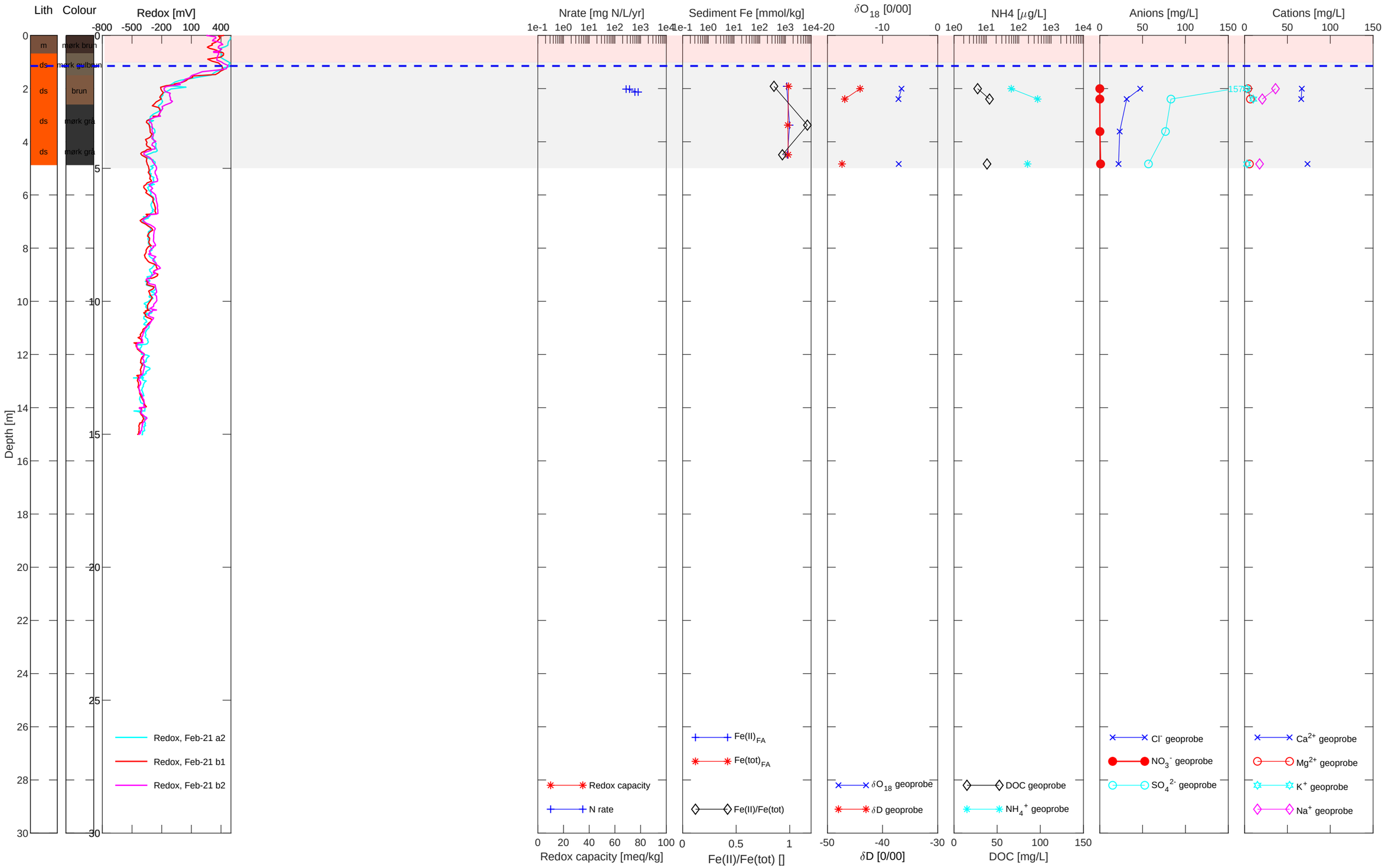
LOOPNr	DGUNr	ID	Depth (m)	mg N/yr	
				repeat1	repeat2
LOOP6	159. 1692	P1	2.31	1127.3	
LOOP6	159. 1692	P1	4.67	2022.0	1648.1
LOOP6	159. 1693	P2	2.02	369.6	273.0
LOOP6	159. 1693	P2	2.13	802.9	601.4
LOOP6	159. 1694	P2a	1.92	1253.8	1186.3
LOOP6	159. 1694	P2a	2.27	448.2	441.2
LOOP6	159. 1695	P3	4.71	404.3	398.3
LOOP6	159. 1695	P3	5.89	853.8	956.0
LOOP6	159. 1695	P3	7.89	502.8	
LOOP6	159. 1697	P5	7.60	179.4	610.6
LOOP6	159. 1697	P5	8.32	558.5	558.8
LOOP6	159. 1697	P5	9.45	491.6	506.1
LOOP6	159. 1698	P6	3.46	828.9	647.6
LOOP6	159. 1698	P6	5.81	52.2	85.1
LOOP6	159. 1699	P7	2.18	4.2	
LOOP6	159. 1699	P7	5.79	1429.4	1038.8
LOOP6	159. 1699	P7	7.25	701.7	694.0
LOOP6	159. 1700	P8	2.09	364.9	357.8
LOOP6	159. 1700	P8	2.28	100.2	136.4
LOOP6	159. 1700	P8	3.39	387.9	365.9
LOOP6	159. 1700	P8	5.83	470.6	455.9
LOOP6	159. 1700	P8	7.00	596.1	615.6

Appendix 4: Well panels illustrating all the collected parameters

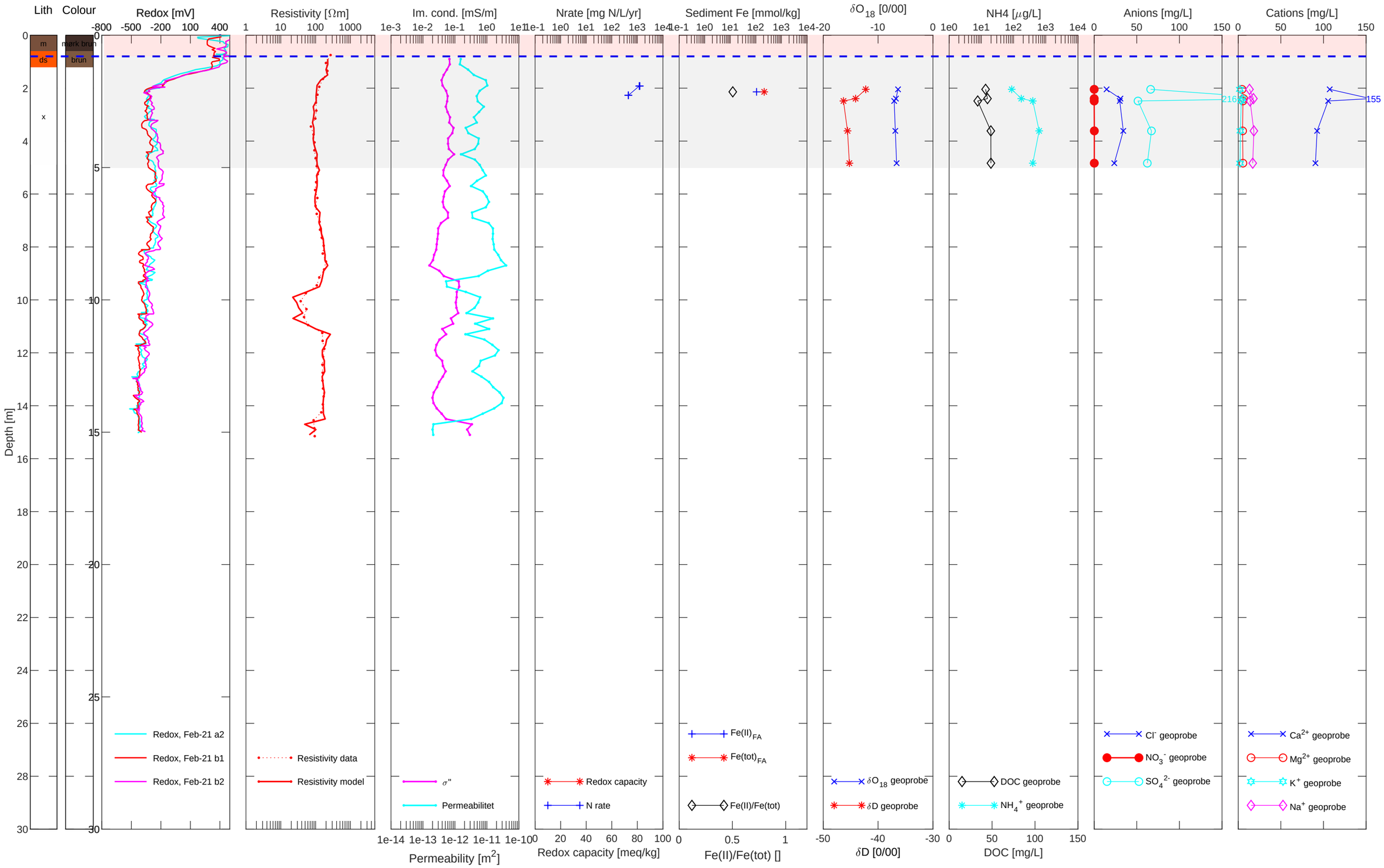
LOOP6 Borehole RX1 , DGUno 159. 1692,



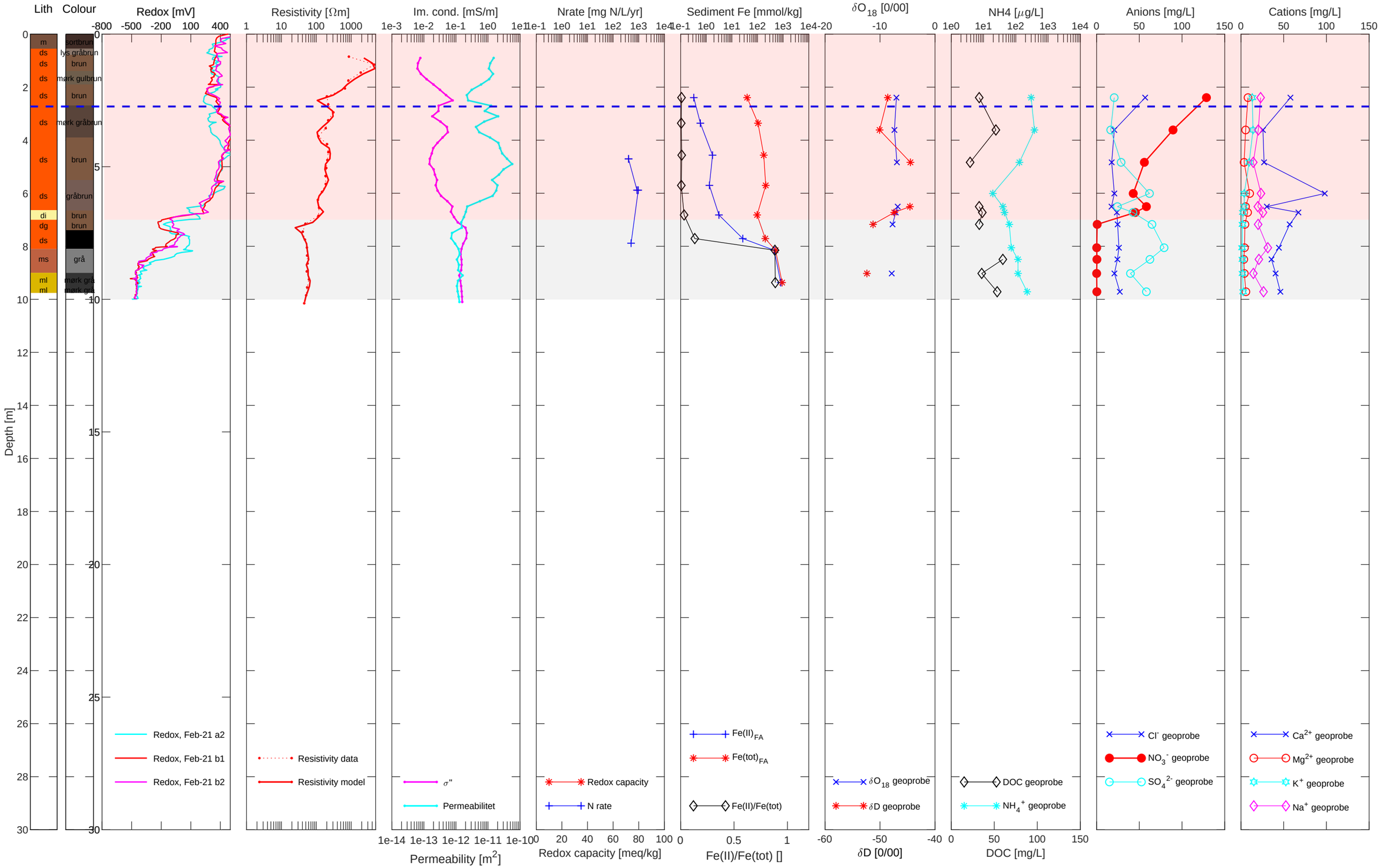
LOOP6 Borehole 2 ; DGUno 159. 1693,



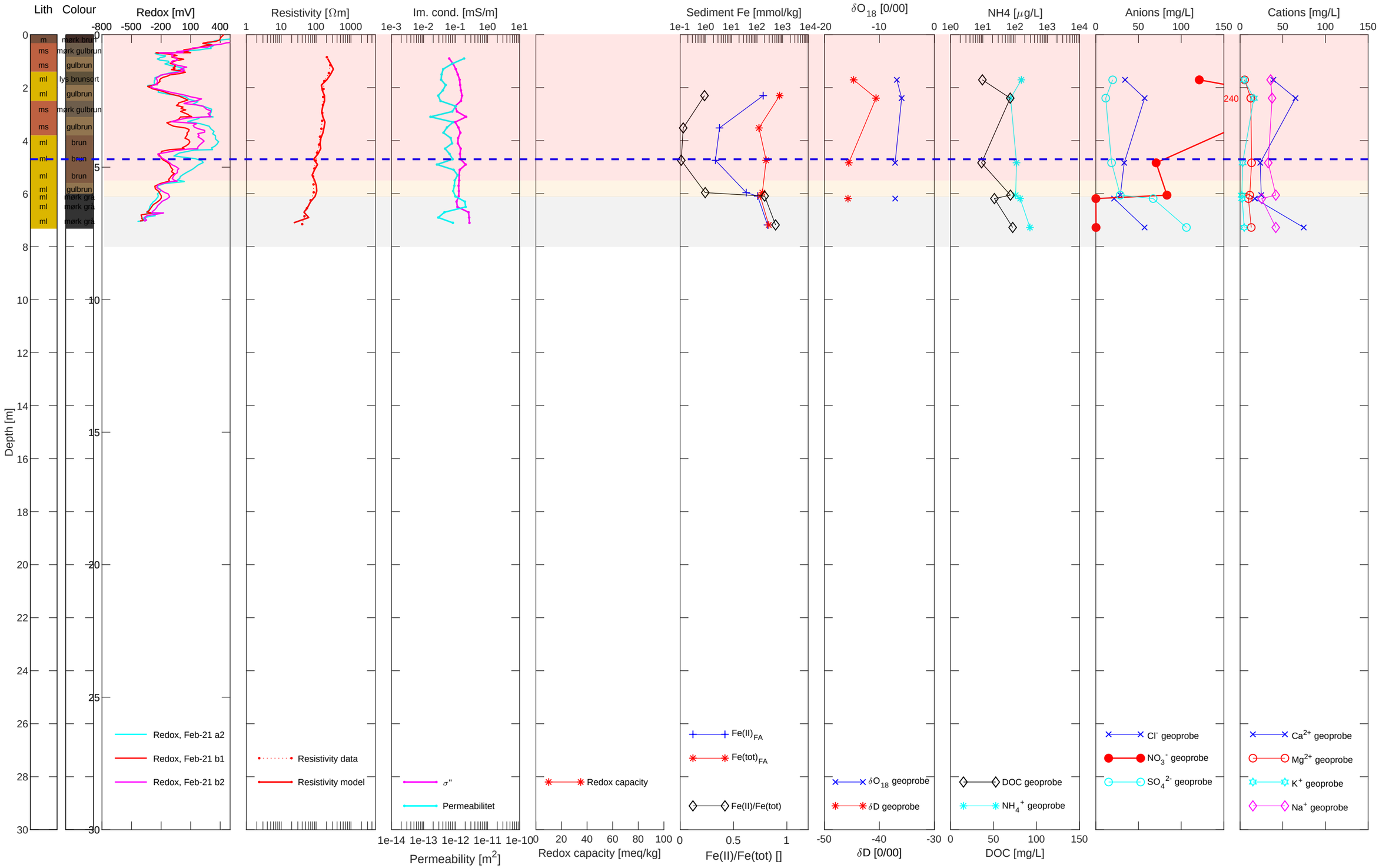
LOOP6 Borehole RX2a , DGUno 159. 1694,



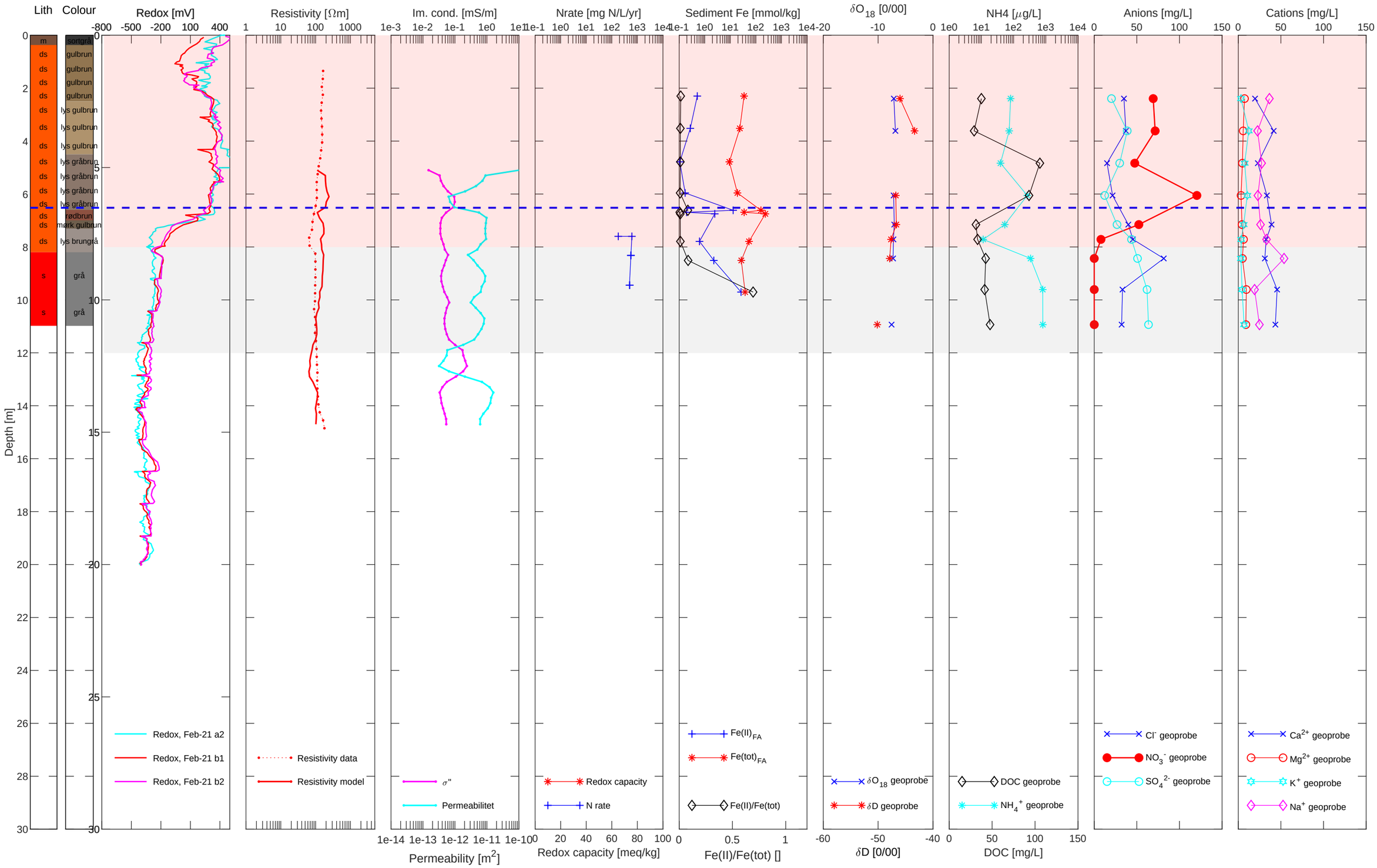
LOOP6 Borehole RX3 , DGUno 159. 1695,



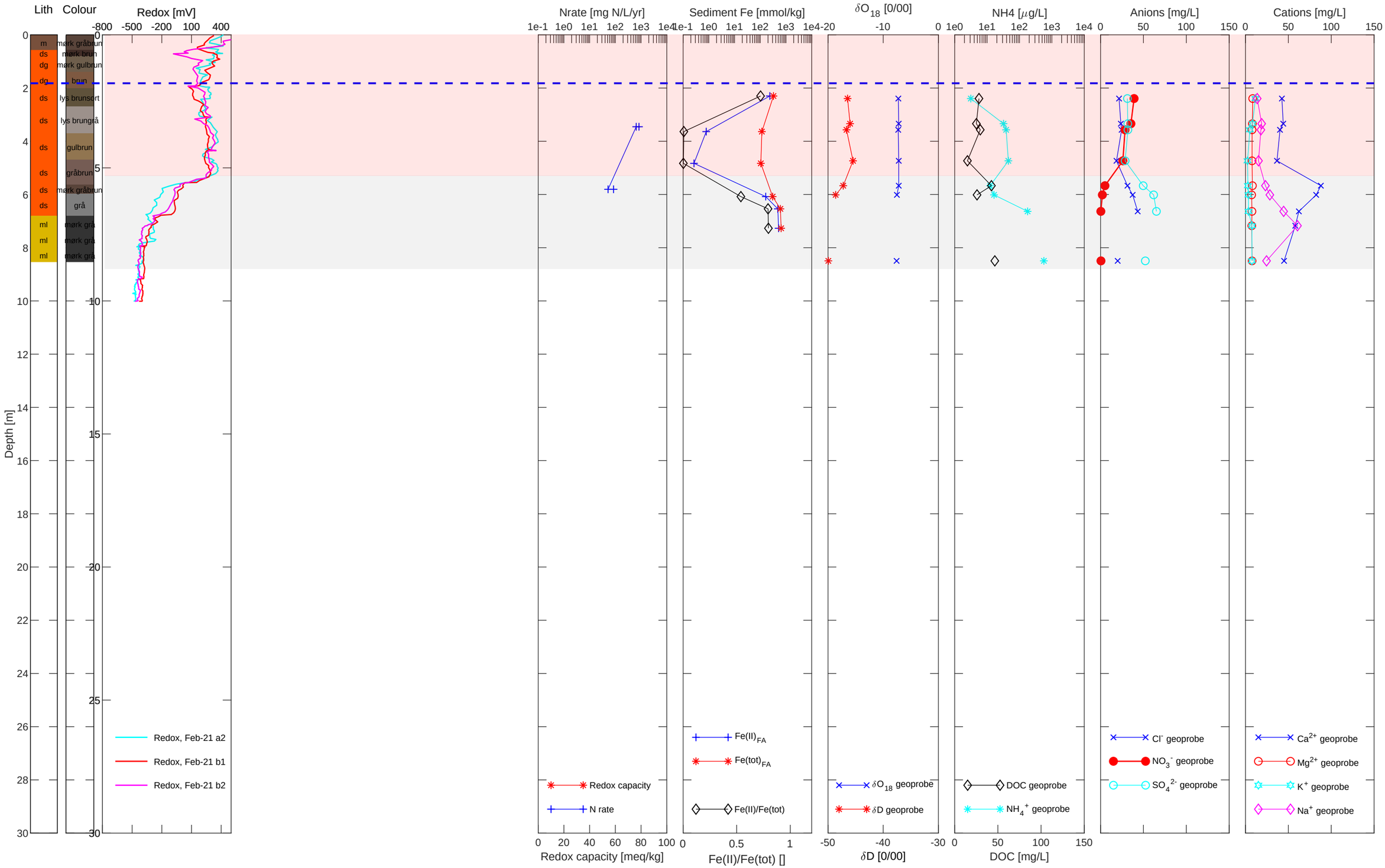
LOOP6 Borehole RX4 , DGUno 159. 1696,



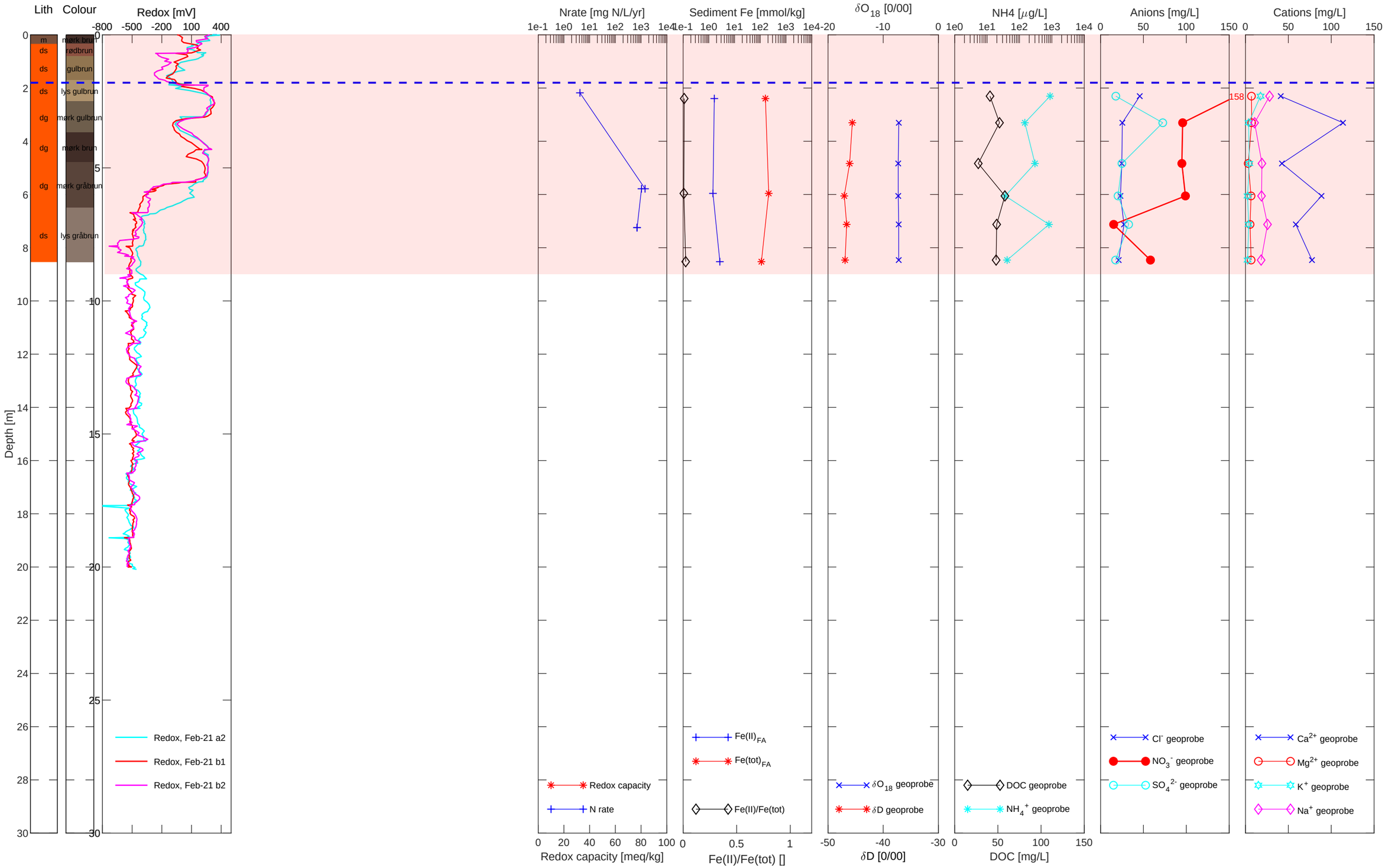
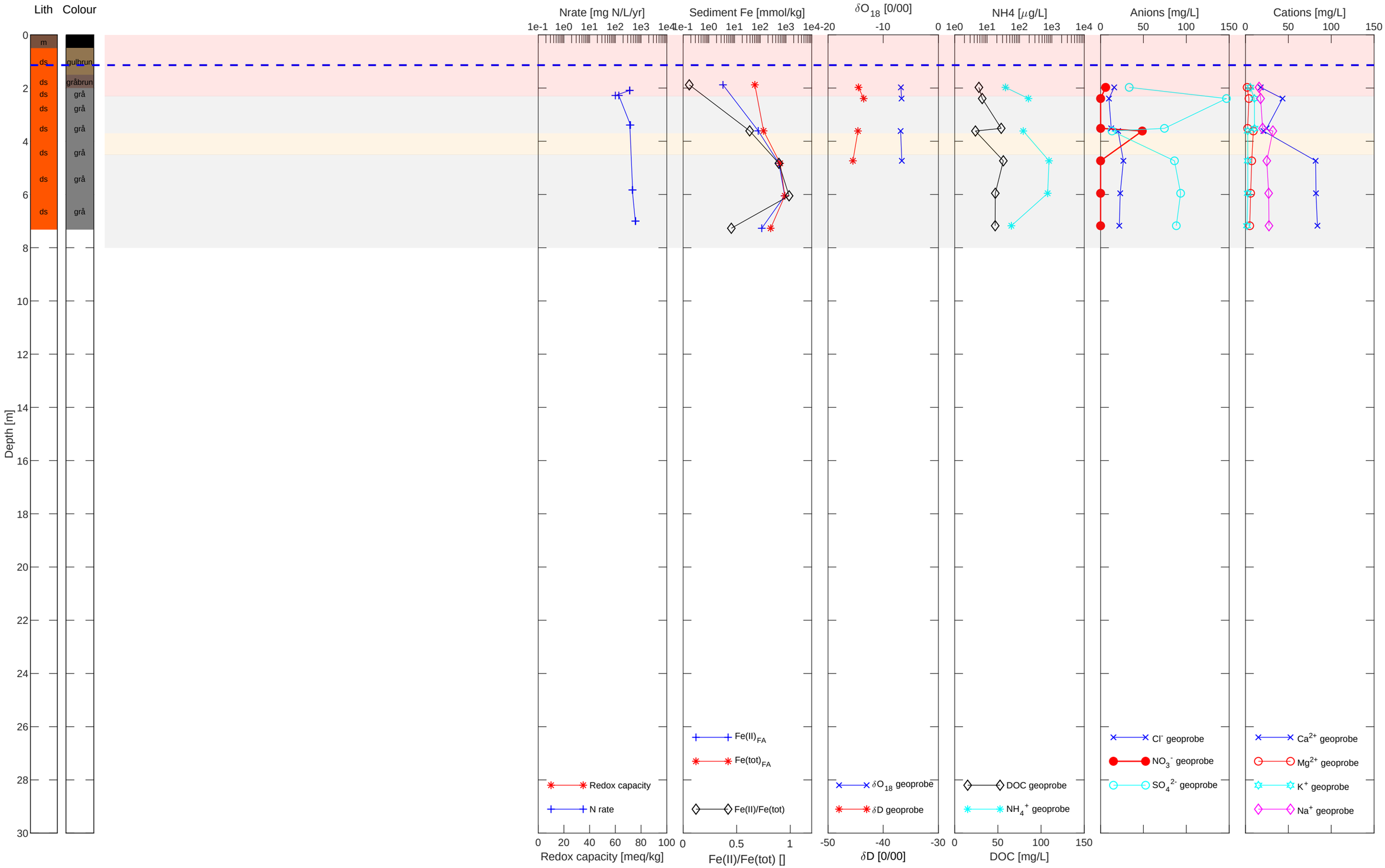
LOOP6 Borehole RX5 , DGUno 159. 1697,



LOOP6 Borehole 6 ; DGUno 159. 1698,



LOOP6 Borehole 7 ; DGUno 159. 1699,



LOOP6 Borehole 8 ; DGUno 159. 1700,